



Pore structure and hydro-mechanical behavior of ryegrass-planted lateritic soil: insights from CT scanning and numerical simulation

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Abstract

Lateritic soils exhibit strong cracking behavior in dry environments, which accelerates rainfall infiltration and weakens soil strength, posing severe risks to slope stability. Plant roots can mitigate these adverse effects, but inappropriate root density may disrupt the soil pore structure and weaken the overall improvement. This study aims to examine the effects of grass roots on the pore structure and hydro-mechanical behavior of lateritic soil. CT scanning was employed to reconstruct the pore structure of soil under varying ryegrass seeding densities. Single-phase flow and finite element simulations were then conducted to evaluate the influence of roots on the hydro-mechanical behavior. The results show that roots reduced porosity and fractal complexity while increasing tortuosity, contributing to a more homogeneous and stable pore network. At a seeding density of 15.0 g/m², the development of cracks and macropores was effectively inhibited, and the porosity and hydraulic conductivity of the soil were decreased. However, excessive seeding density led to localized root aggregation, which increased the heterogeneity of the pore structure as well as the permeability. Meanwhile, roots enhance the strength and deformation resistance of lateritic soil. Overall, during the early growth period, a seeding density of approximately 15.0 g/m² showed the best balance of hydraulic improvement and mechanical reinforcement, particularly during the early growth period. These findings provide a theoretical basis for optimizing seeding density in vegetated soil during the early growth period and demonstrate a CT-driven multi-method framework.

Keywords CT scanning · Grass-planted soil · Hydro-mechanical behavior · Numerical simulation · Seeding density

1 Introduction

Lateritic soil is widely distributed in tropical and subtropical regions [6, 16]. It is characterized by high strength in the dry state but rapidly softens and even disintegrates when wetted, leading to a significant decrease in strength [59]. Due to its high kaolinite content, lateritic soil is also prone to desiccation cracking, which significantly increases its permeability and further enhances its water sensitivity [55]. These characteristics make lateritic soil slopes susceptible to shallow failures under heavy rainfall [15, 61]. To mitigate such risks, various slope protection measures have been adopted, among which vegetation protection has been widely applied owing to its eco-friendly advantage [1, 2, 13, 66]. However, current vegetation protection practices mostly rely on empirical experience, and further research is needed to investigate the influence of plant roots on the hydro-mechanical behavior of lateritic soil. This

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knowledge is essential for the scientific design of slope vegetative protection.

The influence of plant roots on the hydro-mechanical behavior of soil primarily arises from their regulation of the pore structure [12, 24, 32, 34, 45, 56]. In particular, cracks generated under drying or stress conditions can be seen as large interconnected pores that change the overall pore network and greatly weaken the structural integrity. During the growth stage, living roots enhance soil structural stability by secreting exudates and promoting the aggregation of soil particles [12, 25], which decreases both pore volume and connectivity. These structural changes slow down rainfall infiltration, improve water retention, and inhibit preferential flow, resulting in a lower hydraulic conductivity. In contrast, during root senescence or decay, the channels left by decayed roots are often only partially refilled by surrounding soil, forming connected macropores. These macropores serve as preferential flow paths, which significantly increase the local hydraulic conductivity [8, 33]. Regarding mechanical behavior, plant roots interweave with soil particles to form stable root–soil composites, benefiting from their excellent tensile strength and reinforcing effects [24, 40]. This interaction inhibits the formation and expansion of macropores while enhancing soil cohesion and shear strength [38, 47]. Meanwhile, the anchoring effect of roots effectively restrains soil cracking and deformation [5, 36], thereby enhancing soil integrity and further improving its strength [43].

Given the critical role of pore structure in determining the hydro-mechanical behavior of soil, its characterization has gained extensive attention. Commonly used techniques include X-ray computed tomography (CT), scanning electron microscopy (SEM), mercury intrusion porosimetry (MIP), and methylene blue staining [10, 17, 20, 62]. Among these methods, CT scanning provides direct visualization of the spatial morphology and distribution of pores while enabling quantitative measurement of key 3D parameters such as porosity, fractal dimension, and pore diameter. For instance, Wang et al. [50] utilized 4D CT scanning to investigate the microscopic interaction between sand and tall fescue roots, revealing how root growth alters local saturation and pore distribution. Fracica et al. [12] used CT and MIP to investigate the effects of root growth on soil microstructure. They reported that roots modify the pore size distribution by influencing crack formation, filling pores, and promoting soil aggregate formation, thereby altering the hydro-mechanical behavior of soils. Based on CT results, the hydro-mechanical behavior of soils can be analyzed using the pore network model (PNM), lattice Boltzmann method (LBM), finite element method (FEM), or discrete element method (DEM). The PNM simplifies the pore structure into nodes and throats,

allowing soil connectivity and fluid transport efficiency to be quantified through the coordination number of nodes and the tortuosity of throats [3, 27]. Combined with the Hagen–Poiseuille equation and Darcy’s law, the PNM can also be used to efficiently estimate soil permeability [27, 42]. The LBM is a mesoscopic computational fluid dynamics approach that performs seepage simulations directly on a voxel grid and captures details such as dead-end pores and pore bypasses [7, 41, 51]. FEM and DEM are widely applied for mechanical characterization. FEM incorporates pore structure into numerical models to analyze shear damage patterns, plastic zone evolution, and displacement field distribution [23, 44]. DEM explicitly characterizes particle contact behavior and crack propagation, making it more suitable for simulating discontinuous damage and internal crack evolution [46].

Although vegetation has long been recognized for its capacity to improve soil properties, most existing studies have relied on comparisons between vegetated and bare soils, often neglecting the influence of seeding density variations. An increasing body of research indicates that the influence of plant roots on soil behavior is highly sensitive to seeding density [5, 18, 28, 48, 50]. For instance, Zhu and Zhang [66] demonstrated through field investigations that root mass density is a controlling factor in alleviating soil erosion for common grass species. Similarly, Jiang et al. [22] found that increasing root volume density in saturated sand creates a denser network that significantly enhances shear strength and inhibits volumetric deformation. However, these investigations have primarily focused on macroscopic strength or permeability responses of cohesionless or weakly structured soils, whereas the understanding of mesoscopic pore structure evolution under different seeding densities and its implications for lateritic soil hydro-mechanical behavior remains limited. For lateritic soils characterized by an aggregated fabric and pronounced desiccation cracking, the structural response to root growth is expected to be more complex. Consequently, the optimal seeding density capable of simultaneously achieving hydraulic improvement and mechanical reinforcement has not yet been quantitatively determined.

To this end, this study aims to clarify the effects of ryegrass seeding density on the pore structure and hydro-mechanical behavior of lateritic soil. The 3D pore models are reconstructed from CT scan data, and a representative elementary volume (REV) sub-model is selected. Single-phase flow simulations using PNM and LBM are then conducted to compute flow velocity fields and hydraulic conductivity. Furthermore, FEM models incorporating porosity distribution and root reinforcement effects are established to analyze mechanical responses in conjunction with unconfined compression strength (UCS) tests. The

results help identify optimal seeding densities that balance hydraulic improvement and mechanical reinforcement, thus providing a theoretical basis and technical support for the ecological design of slope stability.

2 Materials and methods

2.1 Materials

2.1.1 Lateritic soil

The studied lateritic soil was collected from a construction site (GPS coordinates: 28°08'N, 112°52'E) in Changsha, China (Fig. 1a). After soil retrieval, it was air-dried, decontaminated, crushed, and sieved to obtain particles with sizes less than 2 mm. Subsequently, the basic physical properties of the lateritic soil were tested with reference to the Chinese specification (JTG 3430–2020). The particle grading curves are shown in Fig. 1b, and the measured

physical properties are shown in Table 1. Meanwhile, the soil was analyzed by X-ray diffraction (XRD) and X-ray fluorescence spectrometry (XRF). The results indicate that the primary mineral components of the soil are kaolinite (42.2%), quartz (29.4%), muscovite (20.2%), and hematite (8.0%) (Fig. 1c), while the main chemical components are SiO_2 (57.1%), Al_2O_3 (21.3%), Fe_2O_3 (7.2%), and K_2O (2.5%). The soil is classified as lean clay (CL) according to the Unified Soil Classification System (ASTM, 2017).

2.1.2 Ryegrass

Among various herbaceous plants suitable for slope protection, ryegrass (*Lolium perenne* L.) is widely used in highway slope protection and ecological restoration due to its well-developed roots, rapid growth, and strong adaptability [63, 65]. Its roots are usually branched and rich in fine root hairs, which can combine with soil particles to form a compact root–soil complex in the early growth period, making it particularly effective for shallow slope

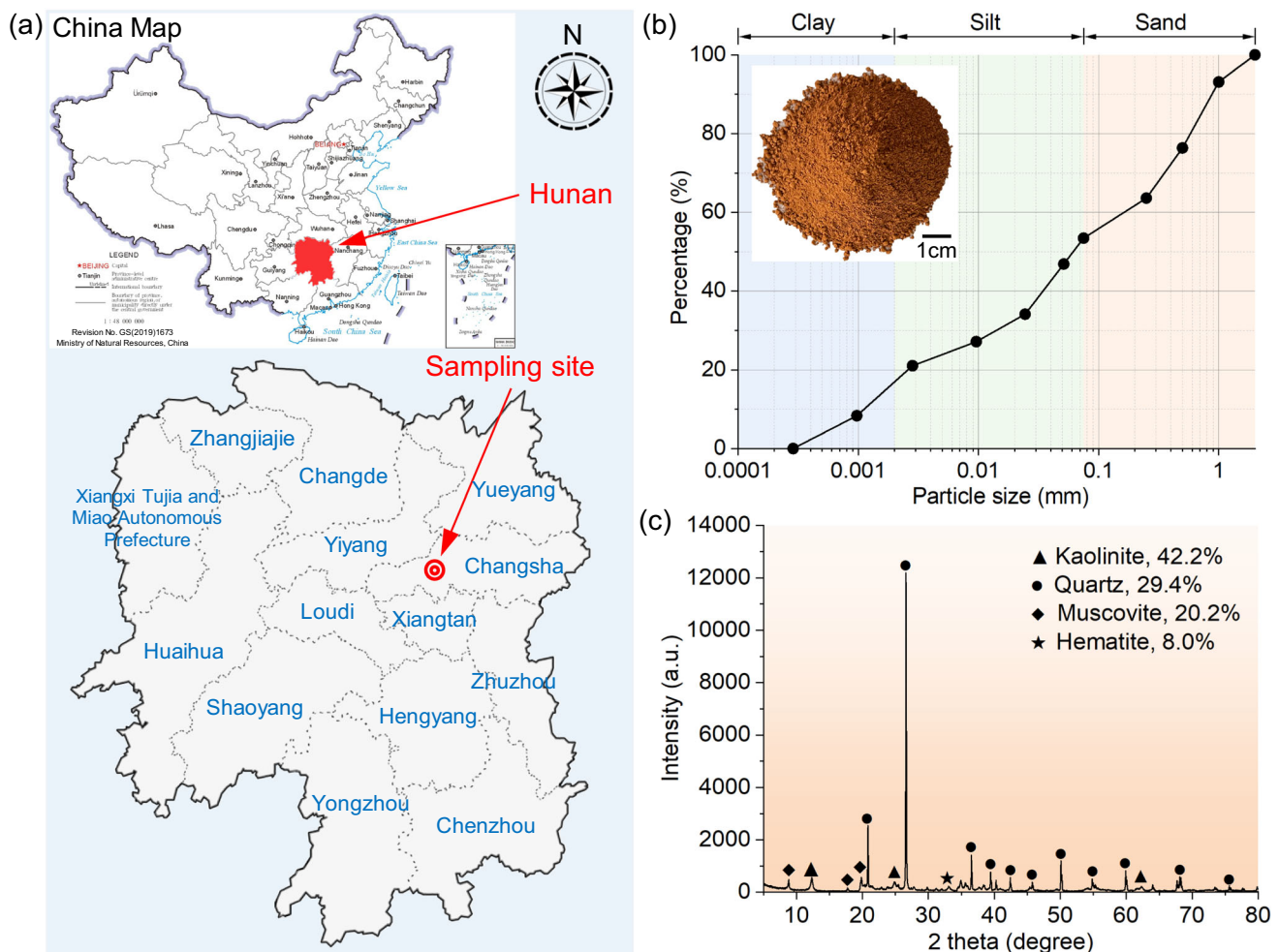
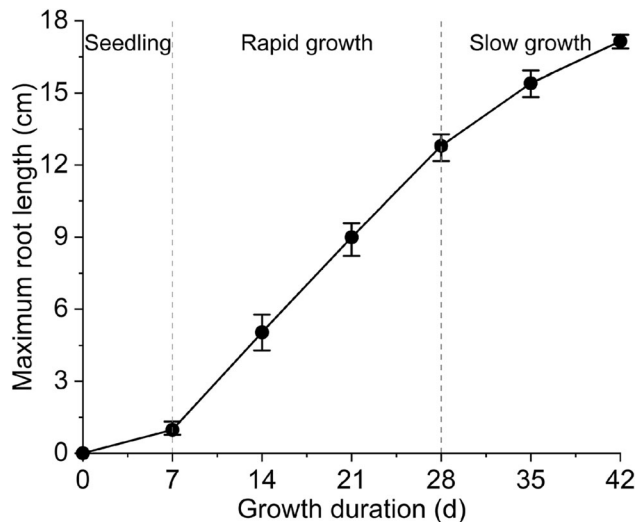


Fig. 1 Lateritic soil: **a** sampling site; **b** particle size distribution curves; **c** XRD pattern

Table 1 Physical properties of the lateritic soil

Specific gravity	Natural dry density (g/cm ³)	Maximum dry density (g/cm ³)	Optimum water content (%)	Liquid limit (%)	Plastic limit (%)	Plasticity index
2.69	1.52	1.78	15.5	40.5	21.2	19.3

**Fig. 2** Relationship between root length and growth duration

stabilization [11, 32]. For this reason, this type of herbage plant was selected for laboratory tests.

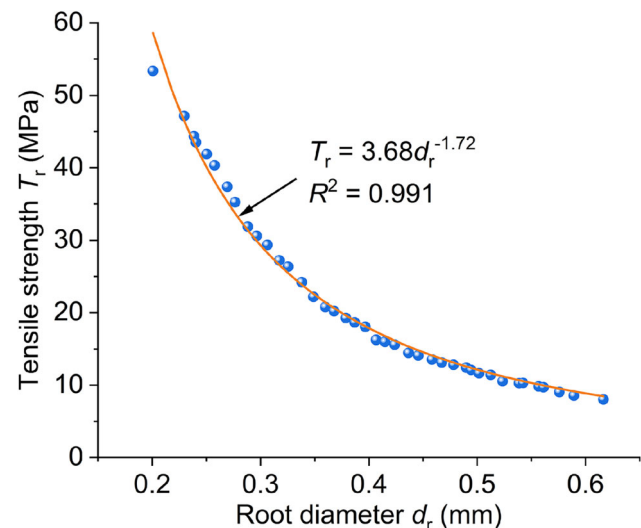
Ryegrass seeds were supplied by China Suqian Fulada Greening Engineering Co., Ltd., with seed purity and germination rates of over 99% and 95%, respectively. The suitable seeding depth for ryegrass is 2–3 cm, and seed germination is typically completed within 7 d after seeding [37]. It was tested and found that the ryegrass roots grew rapidly after germination under the outdoor conditions of spring in Changsha city (average daily temperatures of 10–25 °C), and the growth rate slowed down after about 28 d (Fig. 2). The relationship between root tensile strength and root diameter is shown in Fig. 3, indicating a decreasing trend with increasing root diameter.

2.2 Methods

The laboratory procedures and numerical simulation methods used are shown in Fig. 4, which mainly include grassed specimen preparation, root parameter testing, CT scanning and image processing, hydraulic behavior analysis, and mechanical behavior analysis.

2.2.1 Preparation of grassed specimens

Two sizes of cylindrical specimens were prepared for different tests. For each seeding density, three specimens with a diameter of 50 mm and a height of 100 mm were

**Fig. 3** Tensile strength of ryegrass root

prepared: Two parallel specimens were used for root parameter testing, and one specimen was first subjected to CT scanning and then to UCS testing. This design allowed the pore structure obtained from CT to be directly linked to the subsequent mechanical response of the same specimen. For permeability testing, three parallel specimens with a diameter of 61.8 mm and a height of 40 mm were prepared for each seeding density. Considering that metal molds can cause significant X-ray attenuation effects [4], the 50-mm-diameter specimens were prepared using acrylic molds. All specimens had an initial dry density of 1.52 g/cm³ and an initial water content of 15.5%, corresponding to the natural dry density and optimum water content, respectively. As shown in Fig. 5, seeding spacings of 1 cm, 2 cm, and 3 cm were considered, corresponding to seeding densities (D_p) of approximately 60.0 g/m², 15.0 g/m², and 6.9 g/m², respectively, which cover the seeding density range commonly used in practical engineering. Here, D_p is defined as the total dry mass of seeds divided by the sieving area. A bare specimen ($D_p = 0$ g/m²) and the specimen with a single plant were also prepared for comparison.

The preparation procedure is as follows: (i) The dry lateritic soil was mixed with pure water and sealed for 24 h to ensure uniform water distribution, yielding a target water content of 15.5%; (ii) the wet soil was compacted in layers using the static compaction method to form bare soil specimens [57, 58]; (iii) for grassed specimens, small holes

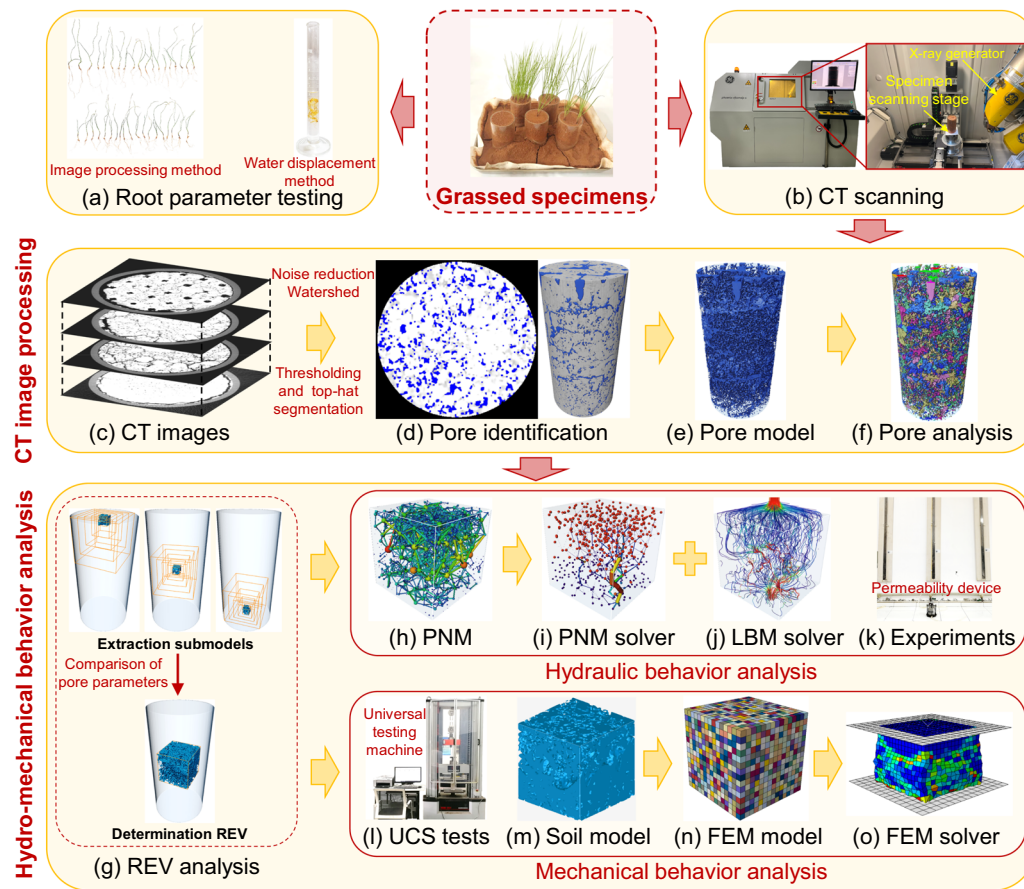


Fig. 4 Laboratory procedures and numerical simulations

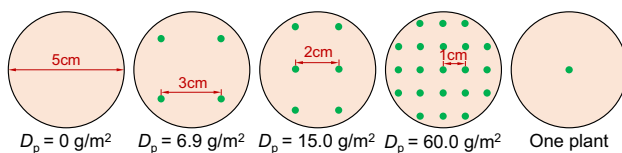


Fig. 5 Seeding scheme

of approximately 1 cm in depth were dug at predetermined intervals on the top of the bare soil specimen, with two ryegrass seeds placed in each hole to enhance germination rates; (iv) the holes were covered with approximately 5-mm-thick dry soil, and water was sprayed uniformly to ensure proper contact between the seeds and surrounding soil.

As shown in Fig. 4, all specimens were placed in trays filled with wet soil approximately 3 cm thick to allow the roots to grow to the bottom of the specimen and continue to extend downward, thereby reducing the influence of the bottom boundary. The trays containing the specimens were placed outdoors (daily average temperature 10–25 °C). For the first three days after seeding, a plastic film was placed over the specimens to provide suitable temperature and humidity conditions for seed germination, after which the

film was removed. During the growing period, the specimens were generally watered twice daily, with approximately 200 mL of water added to each tray at each watering event under normal weather conditions. The watering frequency and volume should be adjusted flexibly according to temperature and rainfall, ensuring the soil remains moist but not visible ponding or overly saturated. Since the ryegrass roots were able to penetrate through the 10 cm specimen about 28 d after seeding (root length of about 12 cm), and the ryegrass was still in the early growth period without aging or decay. Thus, the specimens at 28 d after seeding were considered representative and used for subsequent testing. All specimens were grown for 28 d during the same period (March 18 to April 14, 2024) under the same outdoor environmental conditions and watering management to minimize differences unrelated to seeding density.

2.2.2 Root parameter testing

The roots of each grassed specimen were obtained using the water washing method (Fig. 4a). After photography, ImageJ software was used to process the root images

Table 2 Main pore parameters

Symbol	Parameter	Definition	Equation
P_s	Slice porosity (%)	Ratio of pore area to total slice area	$P_s = \frac{S_p}{S_t}$
P_v	Volumetric porosity (%)	Ratio of pore volume to total specimen volume	$P_v = \frac{V_p}{V_t}$
φ_p	Orientation angle (°)	Angle between pore long axis and horizontal direction	—
SF_p	Shape factor	Deviation of a pore from spherical geometry	$SF_{p^i} = \frac{A_{pi}^3}{36\pi V_{pi}^2}$
AR_p	Aspect ratio	Elongation or flattening degree of a pore	$AR_{p^i} = \frac{L_i}{W_i}$
C_p	Coordination number	Number of connections to adjacent pores	—
T_p	Tortuosity	Tortuosity of the pore channel	$T_p = \frac{L_t}{L_s}$

S_p and S_t are the total area of pores in a certain slice and the area of the slice, respectively; V_p and V_t are the total volume of pores in the 3D model and the total volume of the specimen, respectively; A_{pi} and V_{pi} are the surface area and volume of the i -th pore, respectively; L_i and W_i are the lengths of the long axis and short axis of the pore, respectively; L_t is the actual seepage path length of the fluid in the pore; L_s is the linear distance between the two endpoints of the pore

(binarization, threshold adjustment, root area extraction, and parameter calculation) to acquire the length and diameter of each root. To quantify root mass and volume, the above-ground portion of ryegrass was removed, and the root fresh mass (m_f) was measured. Root volume (V_r) was measured using the water displacement method. The procedure involved submerging the roots in a graduated cylinder containing pure water (graduation: 0.1 ml), with the change in water level representing the root volume [12]. The roots were then immersed in pure water for 48 h, weighed to determine their water-saturated mass (m_s), and subsequently dried at 65 °C for 48 h to determine their dry mass (m_d).

The root–soil dry mass ratio R_d , root–soil wet mass ratio R_w , root bulk density D_{rb} , and water absorption mass ratio δ_r were used to evaluate root content and water absorption capacity. R_d and R_w represent the root mass fraction under dry and natural water content conditions, respectively; D_{rb} is the volume fraction of roots in the soil; and δ_r is the water absorption potential of roots under saturated conditions. The calculation formulas for these parameters are as follows:

$$R_d = \frac{m_d}{M_d} \quad (1)$$

$$R_w = \frac{m_f}{M_w} \quad (2)$$

$$D_{rb} = \frac{V_r}{V} \quad (3)$$

$$\delta_r = \frac{m_s - m_d}{m_d} \times 100\% \quad (4)$$

where M_d and M_w are the dry and wet masses of the soil, respectively (the specimen had a water content of approximately $11 \pm 1\%$ at 28 d after seeding); V is the total volume of the specimen.

2.2.3 CT scanning and image processing

CT scanning was performed using the Phoenix vltomelx s CT scanning system produced by General Electric (GE) (Fig. 4b). The scanning voltage and current were 120 kV and 120 μ A, respectively, with an exposure time of 1000 ms per image, detector binning of 1×1 , and a total of 1000 projection images. Before CT scanning, the part of the grassed specimen was cut off, and all specimens were subjected to the same air-drying treatment at room temperature to reduce the influence of water on image quality and pore identification. The same specimen was used for CT scanning and subsequently UCS testing. Although a large scanning size reduced CT resolution, making roots invisible [50], it captured more macropores and cracks that significantly influence soil hydro-mechanical behavior. At this resolution, direct identification of individual root-induced channels was difficult, and the reconstructed pore phase mainly reflects the overall macropore-crack structure after root growth. Each specimen yielded many slice images (Fig. 4c) with a voxel size of $107 \times 107 \times 107 \mu\text{m}^3$. The boundary effect caused by the mold affected the cracking behavior at the edge of the specimen. In order to retain as much representative internal information as possible, a region with a diameter of 40 mm and a height of 80 mm at the center of the specimen was selected for analysis.

CT image processing was performed using Avizo 3D software, with the main procedures as follows: (i) The 3D model of the specimen was reconstructed by importing the slice images into Avizo and extracting the center region using the volume edit module; (ii) image preprocessing was conducted using the anisotropic diffusion algorithm to reduce image noise; (iii) two materials, namely the soil matrix and air phase, were defined for segmentation; (iv) interactive thresholding was first used to select seed

regions for obvious macropores and the soil matrix; (v) brightness-based segmentation with top-hat enhancement was then applied to supplement seed regions in small pores and fine cracks that could not be clearly identified by thresholding; (vi) the initial thresholds were selected on a representative middle slice and then checked on other slices, with minor manual correction applied when obvious omission or misclassification occurred; (vii) the final pore model was obtained by marker-controlled watershed segmentation, in which the boundaries between pores and the soil matrix were determined automatically from the predefined seed regions (Fig. 4d, e); and (viii) the label analysis module was subsequently employed to calculate the 2D and 3D parameters of the pores (Fig. 4f). Due to the voxel resolution, partial volume effects at pore boundaries could not be completely avoided, but their influence was reduced by focusing on macropores and cracks in the subsequent analysis.

The main pore parameters used in this study are summarized in Table 2. The slice fractal dimension D_s and 3D fractal dimension D_v are used to assess the complexity and heterogeneity of the pore network and are calculated by the box-counting method:

$$D_s = -\lim_{r \rightarrow 0} \frac{\log N(r)}{\log r} \quad (5)$$

where r is the grid side length and $N(r)$ is the number of grids containing at least one pore region. The calculation methods for D_v and D_s are similar, requiring only division and statistics in 3D space.

Both SF_p and AR_p are typically greater than 1, with values closer to 1 indicating that the pores are more rounded. A higher SF_p reflects a greater deviation from spherical geometry, while a higher AR_p indicates that pores are more elongated or flattened. A higher C_p indicates

Table 3 Parameters of the finite element model

Parameters	Seeding density (g/m ²)				
	0	6.9	15.0	60.0	One plant
Elastic modulus E (MPa)	1.5	2.2	3.0	3.3	1.6
Density (kg/m ³)	1687.2				
Poisson's ratio ν	0.49				
Initial cohesion c_0 (MPa)	0.025				
Initial internal friction angle φ_0 (°)	36.9				
Weakening factor χ	3.0				
Root additional cohesion c_r (MPa)	0	0.0041	0.0098	0.0202	0.0018

stronger pore connectivity, whereas a higher T_p indicates a more winding channel with poorer connectivity. In all distribution plots presented in this study, relative frequency is defined on a count basis as the ratio of the number of objects within a given interval to the total number of the corresponding objects in the model.

2.2.4 Numerical and experimental analysis of hydraulic behavior

In numerical simulations, an appropriate REV sub-model is commonly selected to enhance computational efficiency [52]. The maximum side length $L_{\max}$ was determined from the largest square inscribed in the cross section of the cylindrical model. Cubic sub-models with side lengths equal to 20%, 40%, 60%, 80%, and 100% of $L_{\max}$ were then extracted from the top, middle, and bottom regions. For the middle region, the center of each cube coincided

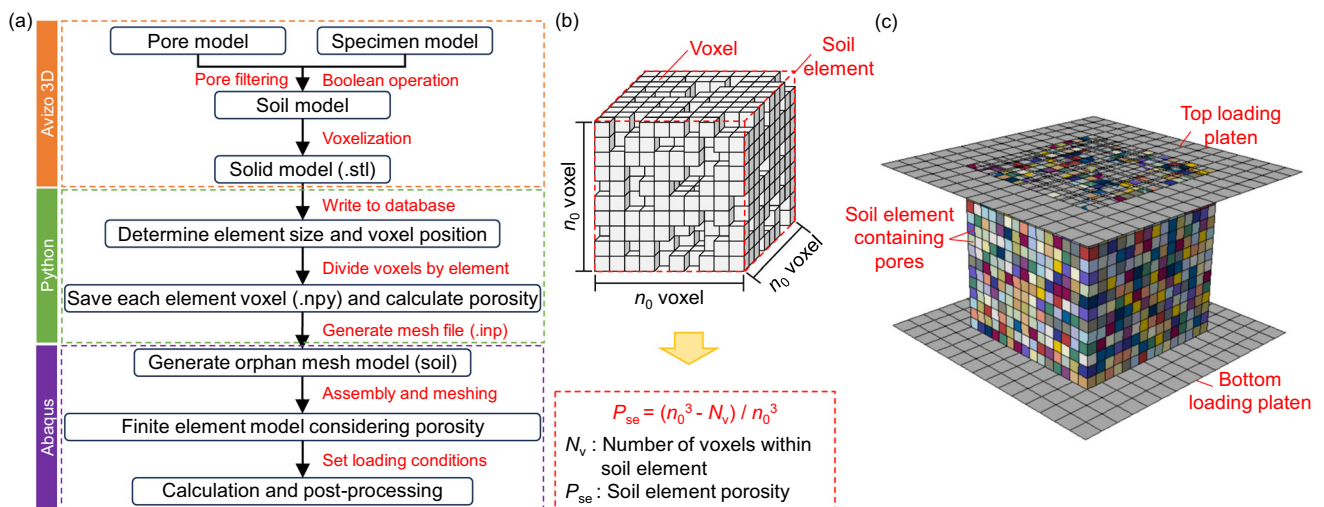


Fig. 6 Finite element numerical simulation method: **a** procedure; **b** element porosity calculation; **c** meshing model

with the center of the cylindrical model. For the top region, the top face of the largest cube was flush with the top surface of the model, and the smaller cubes were defined from the same center point. The bottom region was defined in the same way, with the bottom face of the largest cube flush with the bottom surface of the model. In total, 15 sub-models were obtained. With the bare soil condition as a representative, the pore parameters of these 15 sub-models were calculated and compared with the pore parameters of the complete model, so as to determine the reasonable size and location of the REV (Fig. 4g). The selected REV was the sub-model showing stable pore parameters and the closest agreement with the complete model. After determining the REV, the corresponding pore sub-models were extracted under various conditions.

Subsequently, the REV models were skeletonized for each condition to derive C_p and T_p . Individual pores were then separated to construct an equivalent PNM (Fig. 4h). Finally, single-phase flow simulations were conducted using the PNM method (Fig. 4i) and the LBM (Fig. 4j), with the input pressure and output pressure set at 130 kPa and 100 kPa, respectively, and the fluid viscosity was set to 0.001 Pa·s. In the LBM simulation, the segmented pore space was used directly as the computational domain to solve the saturated steady-state pressure-driven flow field in the z -direction. The convergence criterion was 1×10^{-6} , and the minimum and maximum iteration numbers were 500 and 1×10^6 , respectively.

The saturated hydraulic conductivity was determined using a variable head permeability test. After removing the portion of the grassed specimen above the soil surface, the specimen was vacuum-saturated for 48 h and then placed into a variable head permeability device. The test was

conducted in accordance with the Chinese standard (JTJ 3430–2020) (Fig. 4k). To facilitate comparison between numerical and experimental results, the hydraulic conductivity K_h was calculated from the permeability k_a obtained by single-phase flow simulation using the following equation:

$$K_h = \frac{k_a \rho_w g}{\eta} \quad (6)$$

where ρ_w is the fluid density (1 g/cm^3); η is the dynamic viscosity (0.001 Pa·s); g is the gravitational acceleration (9.8 m/s^2).

2.2.5 Numerical and experimental analysis of mechanical behavior

Mechanical behavior was analyzed through UCS testing and FEM simulations. UCS testing was conducted on a universal testing machine (Fig. 4l) using CT-scanned cylindrical specimens. The test was performed through strain-controlled loading at a rate of $1\% \text{ min}^{-1}$, and the loading was stopped when the stress–strain curve showed a significant peak or when the strain reached 15%.

To further analyze the stress distribution pattern within the soil, a FEM numerical model was constructed based on the CT results and experimental results. Given that a certain degree of segmentation error or noise inevitably occurs during the process of 3D image reconstruction [64], these errors often manifest as small pores or discontinuous pseudo-pores at the voxel level. For lateritic soil, the volume proportion of micropores (pore volume $< 1 \text{ mm}^3$) is less than 30% based on statistical analysis, and it does not change significantly with seeding density. Due to their

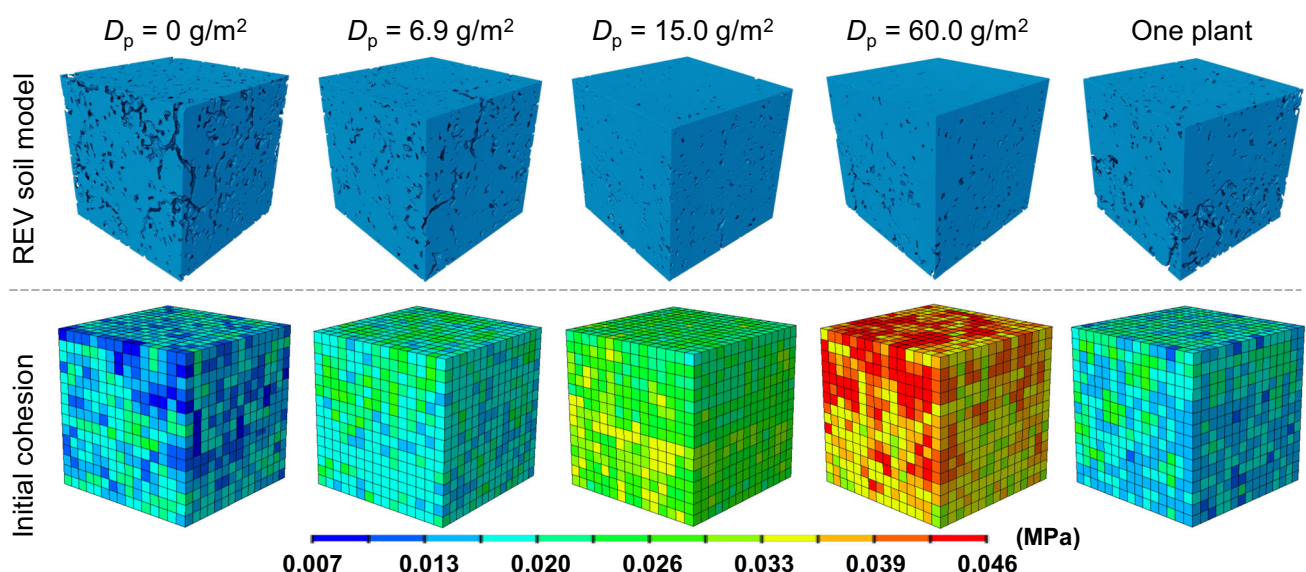


Fig. 7 REV soil models and initial cohesion distribution



Fig. 8 Root images at different seeding densities

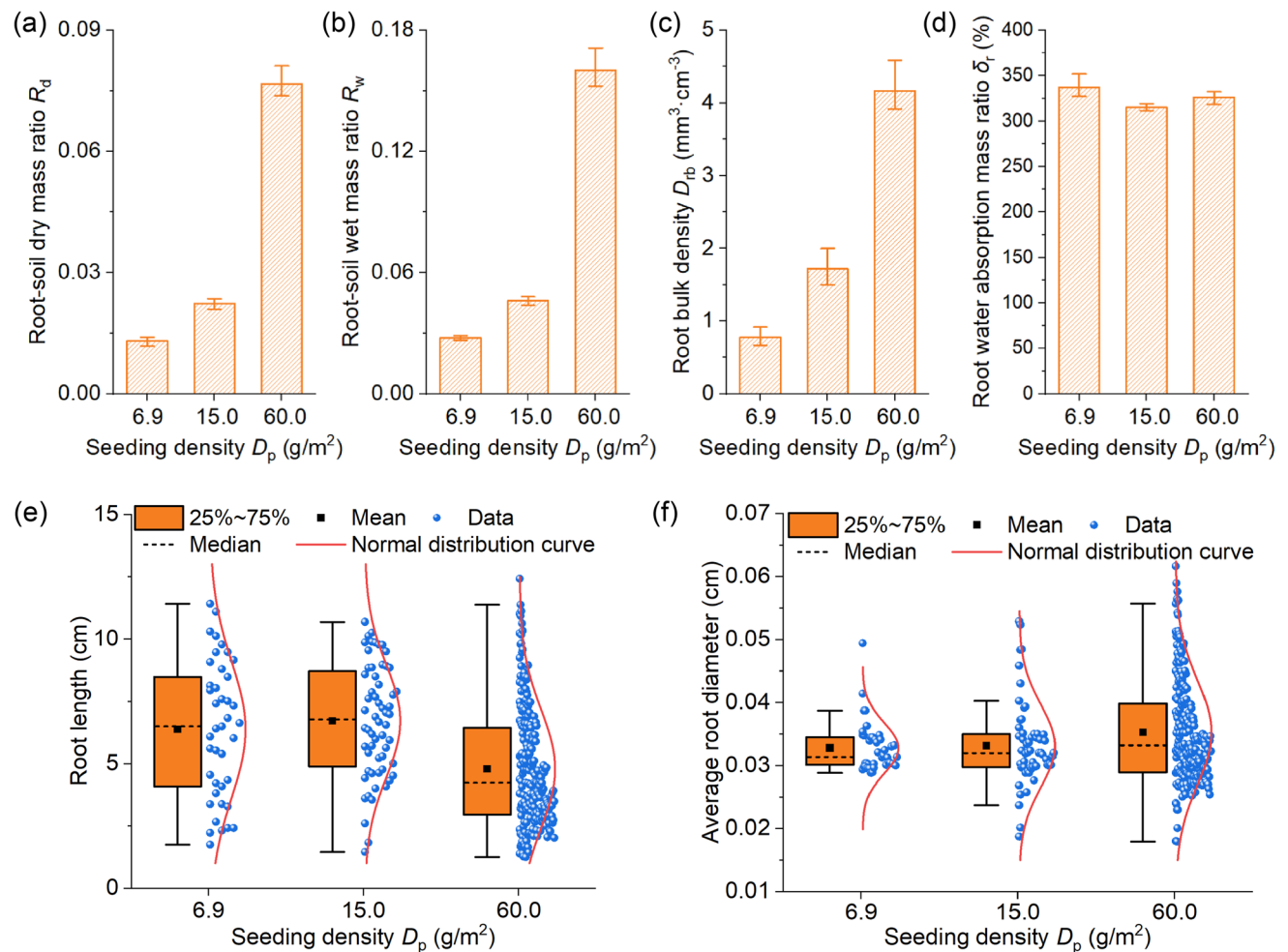


Fig. 9 Root parameters: **a** R_d ; **b** R_w ; **c** D_{tb} ; **d** δ_r ; **e** root length; **f** average root diameter

limited contribution to macroscopic mechanical behavior (e.g., shear strength), micropores were excluded from

further modeling, and pores with volumes greater than 1 mm^3 were selected as the main targets.

As shown in Fig. 6a, the soil model corresponding to the REV pore model (Fig. 4m) was obtained through Boolean operations, and the STL solid model was exported. The voxel information was then written into the database, and the element size and voxel positions were determined. After segmentation, the data were saved as *numpy* files. Each file corresponds to the voxels contained within each element. After counting the number of voxels, the corresponding porosity was calculated (Fig. 6b). To improve computational efficiency, the model was discretized into $15 \times 15 \times 15$ elements, with each element accommodating up to 3375 voxels ($n_0 = 15$). The *numpy* file was converted to an *inp* and imported into the ABAQUS software to create an isolated soil mesh model (Fig. 4n) using the C3D8R element. After assembly and meshing, a FEM

model considering porosity is established.

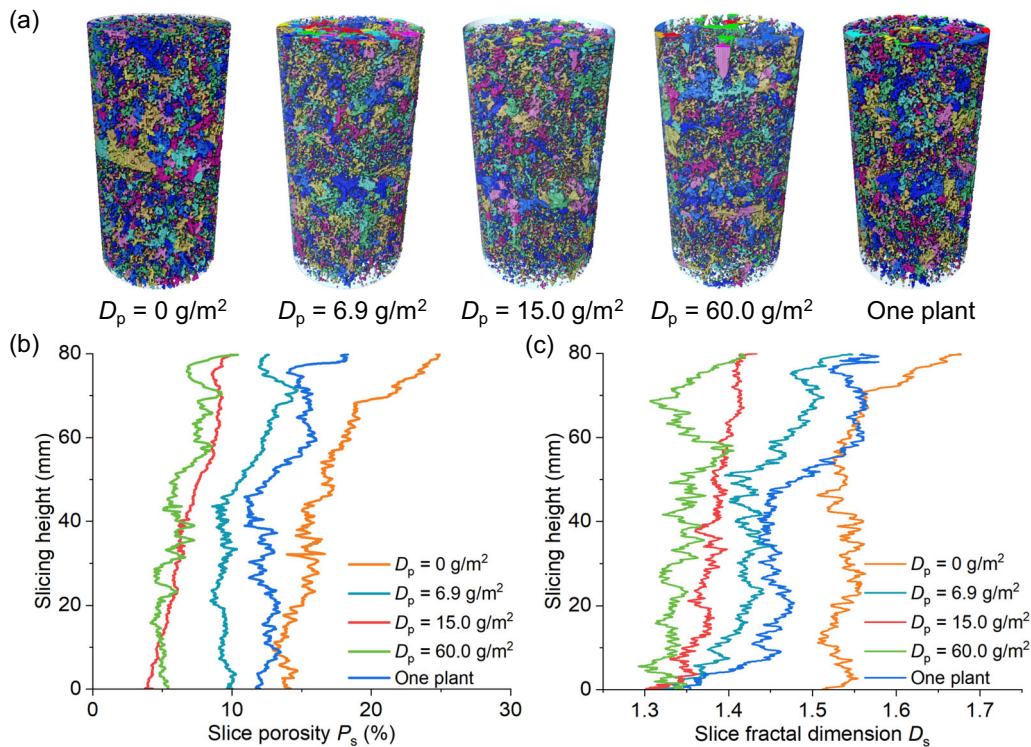


Fig. 10 3D pore models and 2D pore parameters: **a** 3D pore models; **b** P_s ; **c** D_s

Lateritic soil was modeled using the Mohr–Coulomb plasticity model, with post-peak softening accounted for through cohesion decay. The key mechanical parameters of the soil obtained from laboratory tests are shown in Table 3. The initial cohesion and internal friction angle of soil elements under different porosity conditions are calculated according to [15]:

$$c_i = c_0 \cdot \exp(-\chi \cdot P_i) \quad (7)$$

$$\tan \varphi_i = \tan \varphi_0 \cdot \exp(-\chi \cdot P_i) \quad (8)$$

where c_i , φ_i , and P_i are the initial cohesion, initial internal friction angle, and porosity of the i -th element, respectively; P_i was derived from CT reconstruction and corresponds to pores greater than 1 mm^3 ; c_0 and φ_0 are the initial cohesion and initial internal friction angle of the soil at maximum dry density, respectively, for which pores are mostly less than 1 mm^3 , so the corresponding porosity is taken as 0; and χ is an empirical weakening coefficient used to characterize the reduction in c_i and φ_i with

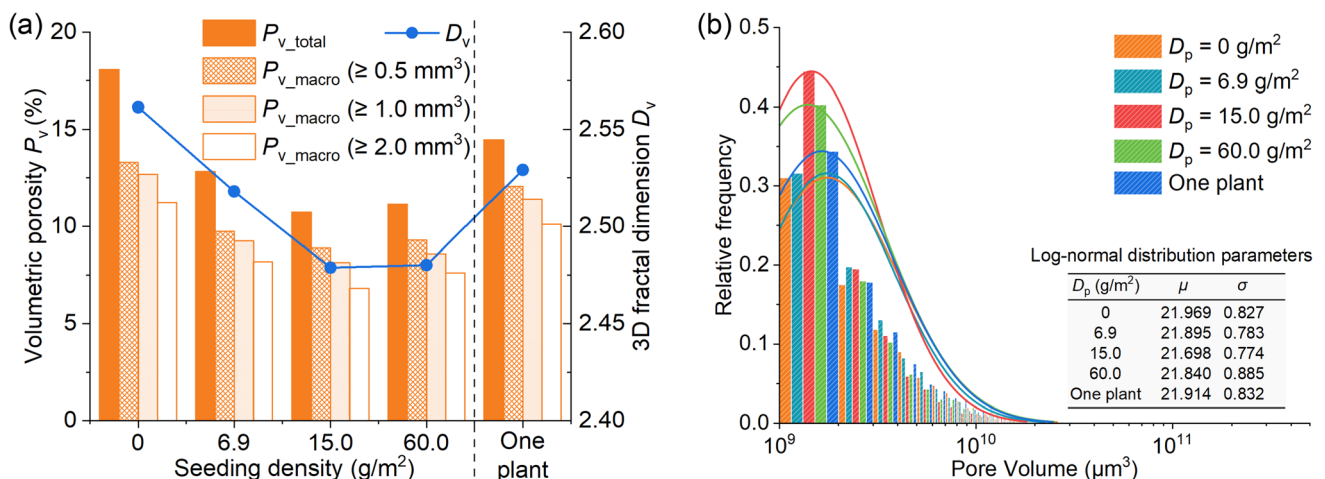


Fig. 11 3D pore parameters of specimens at different seeding densities: **a** P_v and D_v ; **b** pore volume distribution of pores greater than 1 mm^3

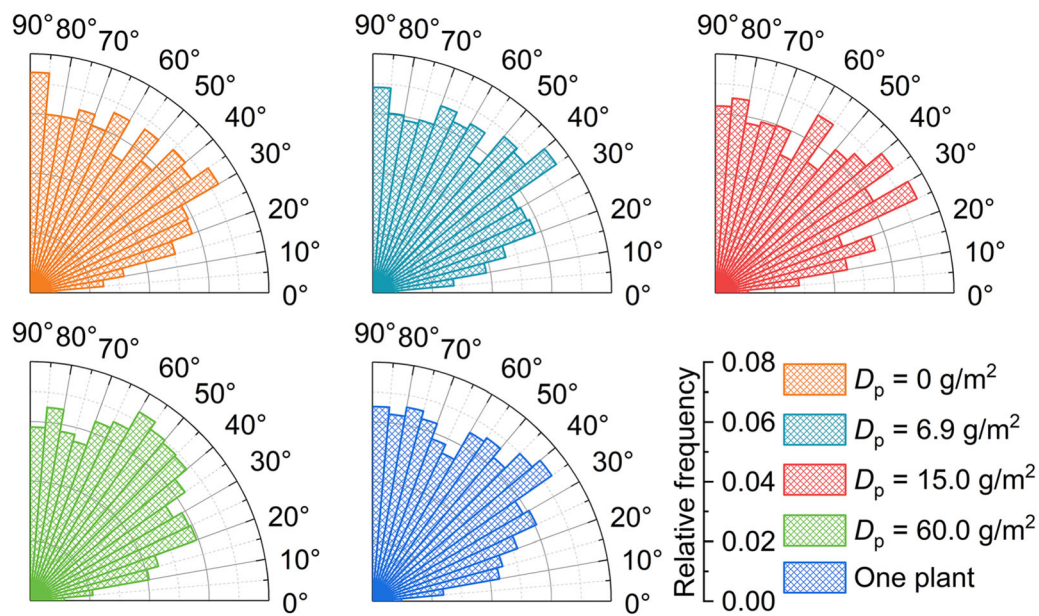


Fig. 12 Spatial distribution of pore orientation angle of specimens at different seeding densities

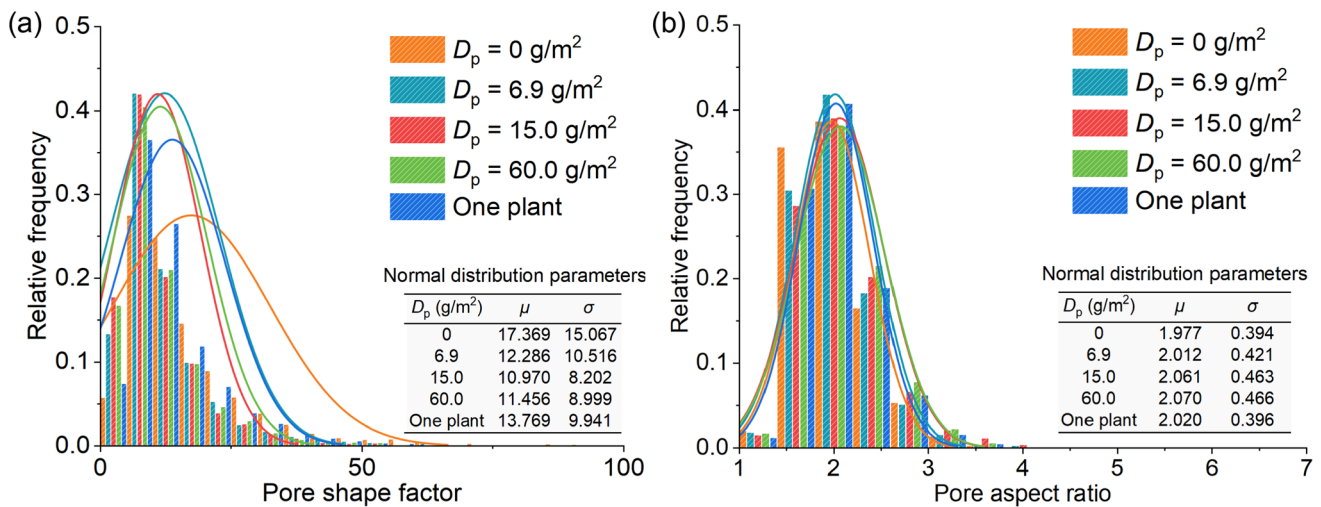


Fig. 13 Pore morphological parameters of specimens at different seeding densities: **a** SF_p ; **b** AR_p

increasing porosity, and its value was selected by comparison with laboratory shear strength test results.

Plant roots can enhance soil strength through functions such as reinforcement and anchoring, as well as the secretion of binding substances. Their contribution to strength can be reflected through the additional cohesion c_r . Based on the Waldron–Wu model [35, 49, 53] and considering empirical corrections, c_r can be expressed as

$$c_r = 0.4 \cdot (\cos \alpha_{rs} + \sin \alpha_{rs} \cdot \tan \varphi) \cdot T_r \cdot RAR \quad (9)$$

where T_r is the tensile strength of a single root, taken as the tensile strength corresponding to the average root diameter value under each condition; RAR is the root cross-sectional area ratio, defined as the ratio of the total root cross-

sectional area to the soil cross-sectional area, and was estimated from the measured root volume obtained by the water washing method; α_{rs} is the angle between the roots and the soil, which is assumed to be 90° in this study. Then, the initial cohesion C_i of the vegetated soil element is as follows:

$$C_i = c_i + c_r \quad (10)$$

In the FEM analysis, c_r was assumed to be spatially uniform within each specimen, while the spatial heterogeneity of the soil was represented by the CT-derived porosity distribution. This heterogeneity is illustrated in Fig. 7, which shows the CT reconstructed REV soil models and the initial cohesion distribution at different seeding

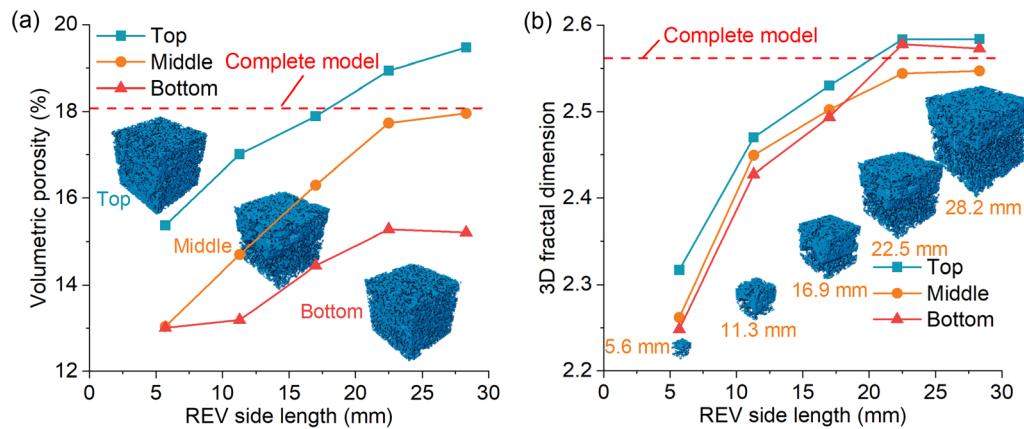


Fig. 14 Pore parameters for REV models with different sizes and regions: **a** P_v ; **b** D_v . Insets in **(a)** show REV models at different regions with the same side length of 22.5 mm, whereas insets in **(b)** show REV models of different side lengths at the middle region

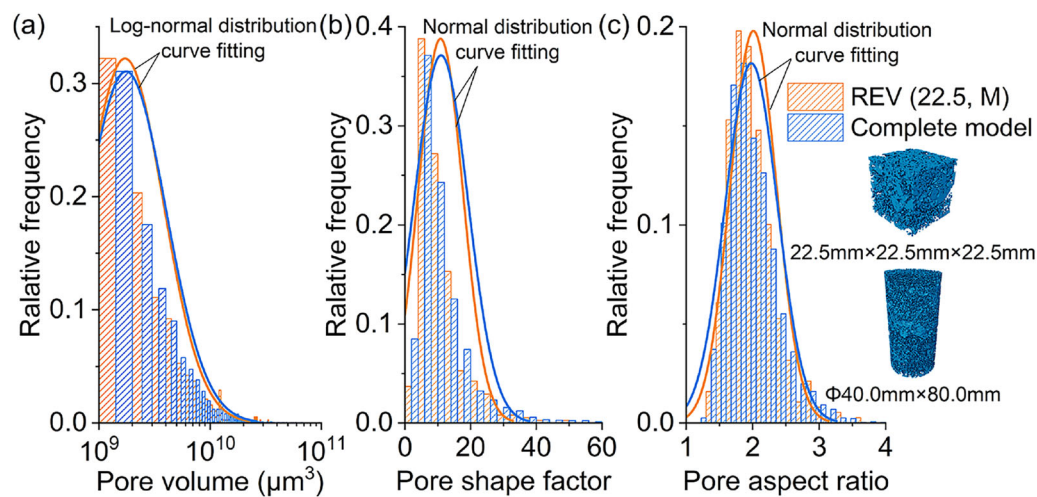


Fig. 15 Comparison of pore parameters between the REV model and the complete model: **a** pore volume; **b** SF_p ; **c** AR_p

densities. Regions with higher porosity correspond to lower cohesion values (blue regions), indicating that the strength parameter distribution in the numerical model reflects the heterogeneity of the pore structure.

The specimen sides were left free, while the top and bottom were placed with rigid loading plates (Fig. 6c). The bottom loading plate was fully constrained, while the top loading plate was free vertically. After numerical calculations and post-processing, the stress–strain curves and stress distribution were obtained (Fig. 4o).

3 Results

3.1 Root parameters

As the seeding density increased from 6.9 to 60.0 g/m^2 , the ryegrass roots became more developed (Fig. 8). The root parameters, including R_d , R_w , and D_{rb} , all showed a

significant increase with increasing seeding density (Fig. 9a–c). δ_r is approximately within the range of 310–350% (Fig. 9d), and the water absorption capacity of the roots is relatively stable at different seeding densities. When the seeding density is 6.9 g/m^2 and 15.0 g/m^2 , the median values of root length and diameter at different seeding densities are close (Fig. 9e, f), approximately 6.5 cm and 0.031 cm, respectively. At seeding density of 60.0 g/m^2 , the median root length is only 4.2 cm, but the maximum root length, median root diameter, and root diameter distribution range are slightly higher than those at the other two densities. At a high seeding density, root competition for water and nutrients is intenser, and most roots are restricted in growth, causing the overall distribution to be concentrated in a shorter range; a few roots extend deeper into the soil to obtain more nutrients and water, forming thicker roots.

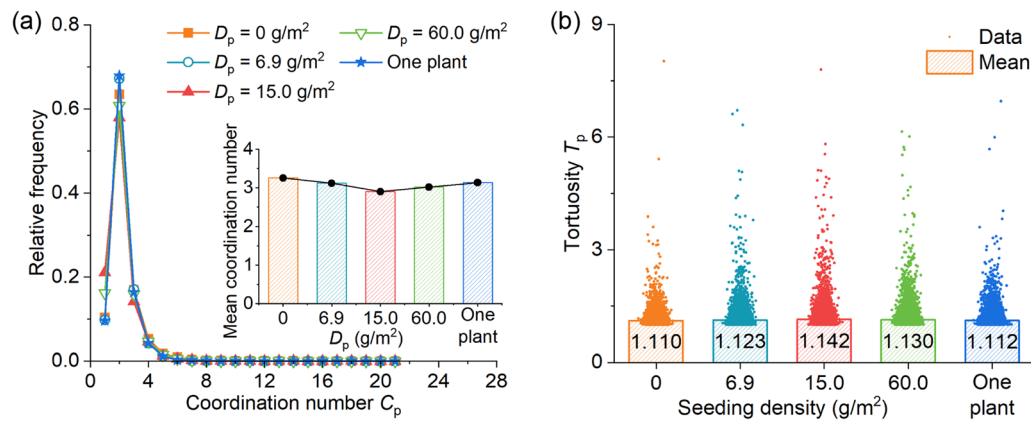


Fig. 16 Connectivity parameters for pore models: **a** coordination number; **b** tortuosity

3.2 Pore structural characteristics

3.2.1 Porosity and fractal dimension

The 3D pore model was reconstructed based on CT scans, and the pore parameters of each slice are shown in Fig. 10. The bare soil specimen and the specimen with a single plant exhibited higher porosity than the other specimens. As shown in Fig. 10b, the P_s generally increased with height from the specimen bottom, as the exposed top experienced alternating wet–dry cycles and rainfall erosion, leading to the formation of numerous cracks, whereas the middle and bottom were relatively less affected by the environment. The P_s of the bare soil specimen ranged between 10 and 25%, while those of the grassed specimens were generally less than 15%. The overall porosity tended to decrease, and the fluctuation range narrowed as the

seeding density increased. However, excessively high seeding density had a limited effect on porosity reduction and might increase the fluctuation range of porosity, resulting in a more dispersed vertical distribution of pores.

As shown in Fig. 10c, the D_s of the bare soil specimen was concentrated between 1.5 and 1.7, with even higher values in the top region, indicating a more complex pore structure characterized by large-scale features such as cracks and macropores. The D_s of grassed specimens was usually less than 1.55. At seeding density of 15.0 g/m² and 60.0 g/m², D_s is even less than 1.4; however, excessive seeding density also causes stronger fluctuations in D_s in the vertical direction.

Figure 11 presents 3D parameters (P_v and D_v) and pore volume distribution curves of specimens at different seeding densities. The overall 3D porosity ($P_{v-total}$) of the bare soil specimen can reach more than 18%, which is

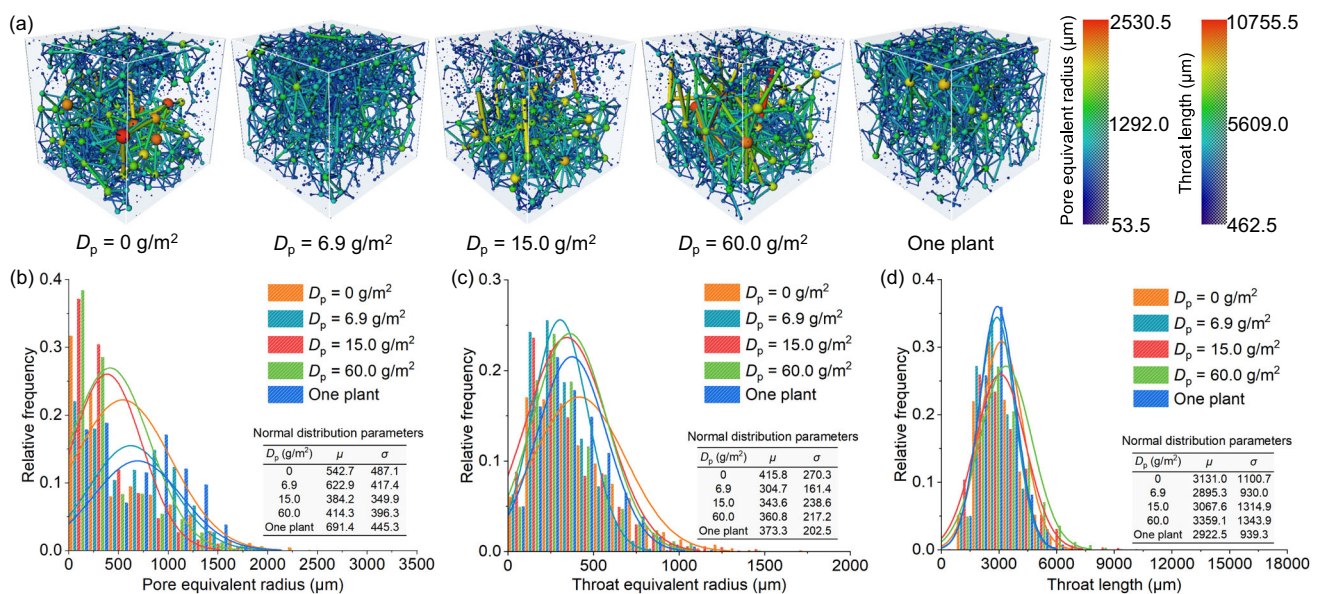


Fig. 17 PNM and parameters of REV at different seeding densities: **a** PNM; **b** equivalent pore radius; **c** equivalent throat radius; **d** throat length

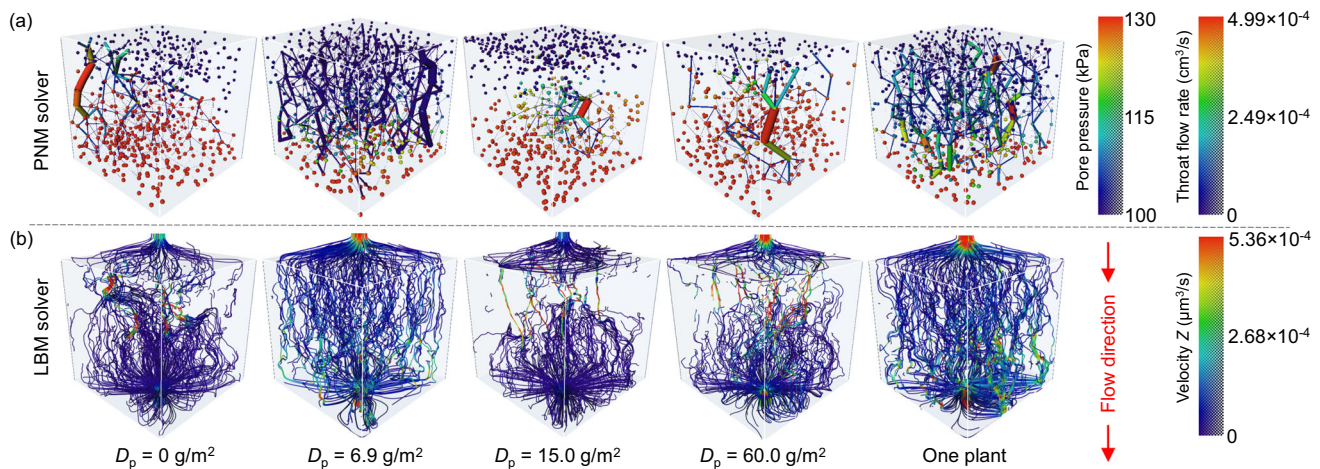


Fig. 18 Flow velocity distribution for single-phase flow simulation: **a** PNM solver; **b** LBM solver

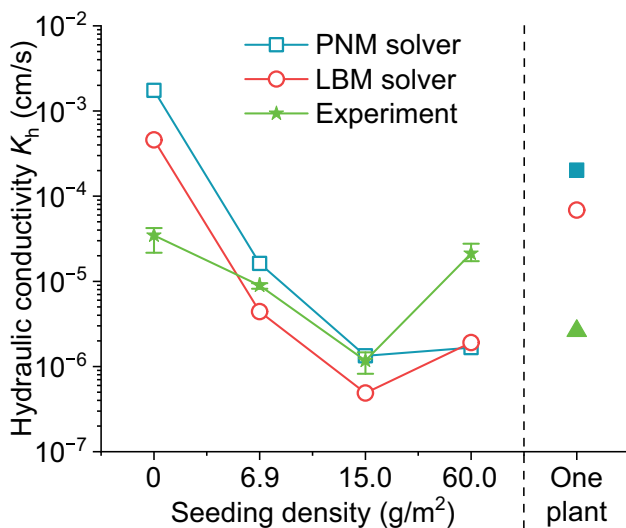


Fig. 19 Hydraulic conductivity of pore model at different seeding densities. Experimental results are plotted as the mean of three parallel tests, and the error bars represent the minimum and maximum values

significantly higher than that of other specimens (Fig. 11a). At seeding density of 15 g/m², the P_{v_total} of the specimens is the lowest, which is about 10.7%. The variation pattern of D_v was consistent with that of porosity, with the bare soil specimen exhibiting the highest value of 2.56. D_v decreased with increasing seeding density, falling to 2.48 at seeding density of 15 g/m². A slight increase in both P_v and D_v was observed at a seeding density of 60 g/m².

To further examine the influence of the macropore definition, the volume fraction of pores larger than 0.5, 1.0, and 2.0 mm³ was also calculated, as shown in Fig. 11a. Although the absolute values of P_{v_macro} vary with the selected threshold, the overall variation pattern remains consistent. Since the main trend remains unchanged, the

threshold of 1.0 mm³ was adopted in the subsequent analysis.

From the pore volume distribution of pores greater than 1 mm³ (Fig. 11b), it is observed that compared with the bare soil specimen, the grassed specimen has more small-volume pores (< 5 mm³), and the proportion gradually increases with increasing seeding density. The pore volume distribution in the grassed specimen is more concentrated, although a few large-volume pores (> 5 mm³) are still present. With a seeding density of 15.0 g/m², the overall distribution curve is more inclined toward small-volume pores, indicating that there are fewer cracks and connecting macropores in the specimens, resulting in a smaller P_v and D_v . In contrast, the percentage of large-volume pores in the bare soil, one plant, and 60 g/m² seeding specimens were relatively high, corresponding to higher porosity and fractal dimension. The parameters of the log-normal distribution fitting indicate that these specimens have greater mean pore volumes and a more dispersed distribution.

It can be summarized that vertical root growth can inhibit the formation of cracks and macropores in the soil, which lowers porosity and promotes pore shape irregularity and vertical pore connectivity. This is primarily because root extension connects some pores along the vertical direction, resulting in smoother transitions in pore structure and reduced overall complexity. Additionally, root growth can compress the surrounding soil, creating voids within the root zone [29]. Due to the CT resolution and the difficulty of distinguishing living roots from the clay matrix, such local features were not quantified separately in this study. At excessively high seeding densities, roots tend to aggregate locally, which increases the irregularity of the pore structure and leads to more significant fluctuations in D_v and in some cases also causes a slight increase in porosity.

3.2.2 Pore orientation angle

Figure 12 shows the distribution of the spatial orientation angle φ_p of pores greater than 1 mm^3 . It can be seen that the percentage of pores in the $0\text{--}20^\circ$ direction is relatively small in all specimens, accounting for less than 15%. In the bare soil specimen, the percentage of pores in the $85\text{--}90^\circ$ and $30\text{--}35^\circ$ directions was the highest, at about 7.3%, while those in the other directions were relatively average, showing obvious anisotropy. This primarily stems from the downward suction gradient formed during the alternating wet–dry cycles experienced by the specimens outdoors. Such a gradient causes internal tensile stresses and consequently promotes the expansion of cracks and macropores in the vertical direction. In the grassed specimens, the percentage of pores in the $85\text{--}90^\circ$ direction decreased, as root penetration downward locally expanded or rotated pores, thus replacing some of the original vertical cracks and macropores. As the seeding density increases, the percentage of pores within the $20\text{--}60^\circ$ range gradually increases, with the highest percentage of pores occurring in the $20\text{--}40^\circ$ range. This indicates that the roots induced new pore channels at a certain depth and direction. However, when the seeding density is 60 g/m^2 , the percentage of pores in the $40\text{--}60^\circ$ direction is higher, and the pore orientation distribution becomes more balanced, resulting in a multi-directional, multi-scale pore, or root development pattern. This suggests that some roots grow vertically or at

a steep angle, causing pores to form and expand over a wider range of angles, thereby increasing pore heterogeneity. The previously observed slight increases in fractal dimensions and porosity at 60 g/m^2 seeding specimens also support the enhanced pore heterogeneity.

3.2.3 Shape factor and aspect ratio

From the pore SF_p distribution (Fig. 13a), it is noted that the relative frequency peak of pore SF_p for the bare soil specimen is 0.275, corresponding to an SF_p of 17.6. Both the mean and standard deviation of SF_p for the bare soil specimen were higher than those of the grassed specimen, indicating a more irregular pore morphology. In contrast, the relative frequency peak of pore SF_p for the grassed specimen is greater than that of the bare soil specimen, while the corresponding SF_p is smaller. The mean and standard deviation of pore SF_p gradually decrease with increasing seeding density, but slightly increase under high-density seeding ($D_p = 60 \text{ g/m}^2$). During the root growth process, the pores are connected or expanded into more rounded shapes, with moderate morphological irregularity, rather than irregular macropores.

For pore AR_p (Fig. 13b), the mean and standard deviation of each specimen did not change significantly, with the mean concentrated between 1.977 and 2.031 and the standard deviation between 0.394 and 0.466. The

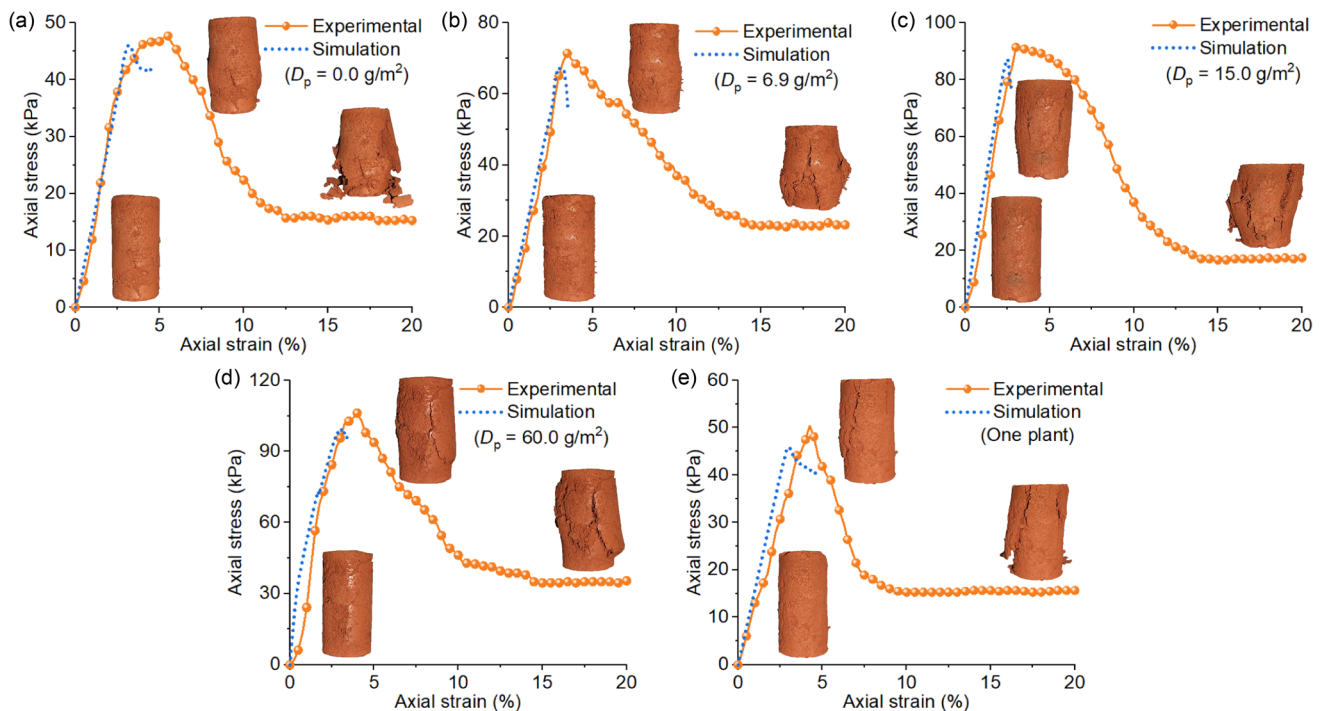


Fig. 20 Stress–strain curves obtained from UCS experiments and numerical simulations: **a** $D_p = 0 \text{ g/m}^2$; **b** $D_p = 6.9 \text{ g/m}^2$; **c** $D_p = 15.0 \text{ g/m}^2$; **d** $D_p = 60.0 \text{ g/m}^2$; **e** one plant

differences among specimens are small, and no obvious trend is observed with seeding density.

Combining the two indicators, the bare soil specimen mainly had large cracks and irregular macropores with higher SF_p . After grassing, the soil pores tended to become more rounded, as indicated by the decrease in SF_p . In contrast, the changes in AR_p were small among different seeding densities, suggesting that vegetation mainly affected pore irregularity rather than pore elongation or flattening.

3.3 REV analysis

To determine the appropriate REV, cubic sub-models of different regions and sizes were extracted from the pore model of the bare soil specimen, and their P_v and D_v values were calculated, as shown in Fig. 14. Both P_v and D_v of the sub-models initially increased and then gradually stabilized, reaching steady values when the cube side length was 22.5 mm. This indicates that a side length of 22.5 mm is sufficient for the cubic sub-model to represent the pore structure of the complete model. The pore parameters of sub-models at different regions also exhibit differences, with P_v and D_v values in the top region being higher than those in the middle and bottom regions. Among the cubic sub-models with side lengths greater than 22.5 mm, the P_v and D_v of the middle sub-model are closest to those of the complete model. Therefore, a cubic sub-model with a side length of 22.5 mm from the middle region was selected as the REV.

Figure 15 compares the distributions of pore volume, SF_p , and AR_p between the selected REV and the complete model to validate its representativeness. In the pore volume distribution (Fig. 15a), the curves of the REV and the complete model both exhibit a right-skewed pattern, indicating that most pore volumes are concentrated in a

relatively smaller range (10^9 – $10^{10} \mu\text{m}^3$). The two curves are relatively close, suggesting that the REV can effectively reproduce the pore volume distribution in the complete model. Furthermore, the REV and the complete model are generally consistent in the peak ranges and distribution patterns of SF_p (Fig. 15b) and AR_p (Fig. 15c). Thus, the selected REV (side length of 22.5 mm, middle) is suitable for representing the complete model in subsequent numerical simulations.

3.4 Pore connectivity parameters

Figure 16 shows the relative frequency and mean values of C_p and T_p . The C_p of the pore nodes in each specimen was mainly distributed between 1 and 4 (Fig. 16a), indicating that most pore nodes were connected to only a few throats, with only a small portion showing strong connectivity. Among them, the percentage of pore nodes with a high coordination number and the mean C_p of the bare soil specimen were higher than those of the grassed specimens. This phenomenon is likely related to the presence of large cracks or macropores in the soil, which made the C_p of local pores much higher than that of others. After grass planting, the mean C_p of pores decreased, with the lowest mean C_p observed at 15.0 g/m^2 . The reason is that the roots connect some pores vertically, dividing the original crack network into smaller isolated pore groups and thus limiting the horizontal connectivity of many nodes. However, the mean C_p increases when the seeding density is raised to 60.0 g/m^2 . This indicates that in an excessively dense root environment, roots are more likely to intersect and reorganize existing pore structures, thereby forming larger macropores and providing more connection paths for local pore nodes.

The T_p of throats in all specimens was mainly concentrated between 1 and 3, with mean values ranging from

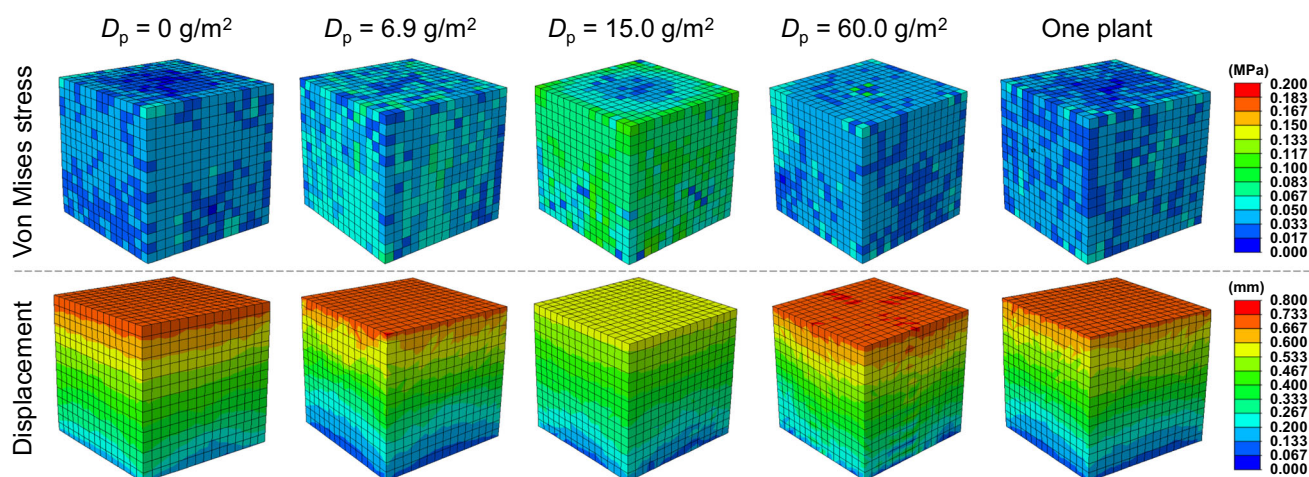


Fig. 21 Stress and displacement distribution of the numerical model at maximum axial stress

1.11 to 1.15 (Fig. 16b). The mean T_p of the bare soil specimen was the lowest, showing that most of the throats were connected and had low tortuosity. The connectivity between high coordination number nodes largely depends on large cracks or macropores, which are often connected through shorter and straighter throats. After grass planting, the mean T_p increased, reflecting a more tortuous internal pore network. This may be attributed to the squeezing and filling effects of the roots, which often increase the tortuosity of the seepage paths, resulting in poorer pore connectivity.

For a more intuitive description of the spatial distribution of pore nodes and throats, the PNM was constructed and its parameters were statistically analyzed (Fig. 17). Under the bare soil and 60.0 g/m² seeding conditions, the PNM contains several macropores connected to long throats. The pore and throat sizes are more frequently distributed in the large radius range (> 800 µm) (Fig. 17b,c), and throat lengths are mainly concentrated between 1000 and 6000 µm, peaking at 2000–4000 µm, reflecting the development of large cracks and macropores. In other conditions, although the total pore count is higher, the pores are primarily of medium to small sizes and are more evenly distributed, forming a relatively uniform pore network. In addition, grassed specimens with moderate to high seeding densities (15.0 and 60.0 g/m²) showed a higher percentage in the long throat range (> 5000 µm) (Fig. 17d), suggesting that the roots could form deeper channels in some parts, which might positively influence the infiltration process.

3.5 Hydraulic behavior analysis

The velocity fields obtained from single-phase flow simulations using the PNM and LBM are shown in Fig. 18, and the corresponding hydraulic conductivity is presented in Fig. 19. From the PNM results, some high-pressure nodes and high-velocity seepage paths were observed under the bare soil and 60.0 g/m² seeding conditions. This is due to the dominance of large cracks and macropores as the main seepage paths, which enable fluids to flow rapidly along concentrated paths. Under the 6.9 g/m² seeding conditions, the pressure distribution is more dispersed, with flow in the throats diverted into multiple small branches, indicating that the pore network is subdivided into smaller sub-networks. At 15.0 g/m², only a few penetrating cracks and macropores forming high-velocity seepage paths appeared in the central region, while the flow velocity in the remaining regions was extremely low, resulting in an increase in overall tortuosity (Fig. 16b).

Under one plant seeding condition, relatively uniform pressure and seepage path distribution were observed, but several distinct high-velocity seepage paths were

developed in the vertical direction. The LBM solution results (Fig. 18b) are generally consistent with the PNM results regarding flow distribution. The flow paths near the central parts of the top and bottom boundaries are more concentrated. This is because the connected pore openings on the truncated top and bottom faces are distributed non-uniformly, and the main through-going channels in the z -direction are mainly concentrated in the central region. In the bare soil specimen, the high-velocity seepage paths were fewer and more concentrated, while in the 60.0 g/m² grassed specimen, they were more numerous and widely distributed, reflecting a more complex pore network. A relatively uniform distribution of seepage paths was observed in the one plant and 6.9 g/m² grassed specimens, with a large number of branch seepage paths with intermediate flow velocities appearing in the lower region. In the 15.0 g/m² grassed specimen, the seepage paths were mainly concentrated in the lower region with low flow velocities, and a few high-velocity seepage paths were observed in the crack regions.

Figure 19 shows that the bare soil specimen exhibited the highest hydraulic conductivity in both numerical solutions, due to the relatively greater abundance of large cracks and macropores that acted as the main seepage paths. As the seeding density gradually increased, the hydraulic conductivity decreased significantly because the roots divided the soil macropores into smaller pore sub-networks. The hydraulic conductivity was lowest when the seeding density was 15.0 g/m². However, as the seeding density continued to increase to 60.0 g/m², both the hydraulic conductivity values obtained from numerical simulations and laboratory tests increased, likely because dense root aggregation reopened or expanded some existing cracks and pores. Although the specimen with a single plant formed obvious vertical seepage paths locally, the spatial extent was relatively limited, making its hydraulic conductivity slightly lower than that of the bare specimen.

Additionally, since the LBM captures more local flow details, while the PNM emphasizes the main seepage paths and simplifies some microscopic energy dissipation, the hydraulic conductivity calculated by LBM is slightly lower than that calculated by PNM. The variation patterns of numerical simulations and laboratory tests are generally consistent, though differences in magnitude exist, which may be attributed to the resolution limits of CT scanning and the complexity of root–soil interface effects. Overall, at low and moderate seeding densities, root growth tends to partition the pore network into finer and more tortuous paths, thereby disrupting the continuity of macropores and significantly reducing overall permeability. At one plant and high seeding densities, macropores and their connecting paths are more prominent, enabling the formation of

multiple high-velocity seepage paths, which leads to a smaller decrease in hydraulic conductivity.

3.6 Mechanical behavior analysis

The stress–strain curves of specimens at different seeding densities were obtained through UCS tests and numerical simulations (Fig. 20). The test results indicate that all specimens exhibited pronounced strain-softening behavior, with axial stress increasing rapidly to a peak, followed by a post-peak decline and eventual stabilization. The UCS of the bare soil specimen was the lowest, at only 47.7 kPa, primarily due to its loose structure and relatively high porosity, which provide minimal resistance to axial compression [14, 59]. As the seeding density increased, the UCS gradually increased to reach 106 kPa at a seeding density of 60 g/m², showing an increase of over 120%. This trend does not strictly correlate with the variation in porosity (Fig. 11a), suggesting that roots not only influence strength by altering the pore structure, but may also provide additional cohesion and tensile strength through mechanical reinforcement and enhanced matric suction [30, 39].

Additionally, roots can enhance the soil's resistance to deformation, causing the peak stress to occur at a smaller axial strain. This phenomenon becomes more pronounced with increasing seeding density. The morphological changes also reveal that the bare soil specimen exhibits expansive deformation and significant fragmentation upon failure. In contrast, grassed specimens show lower deformation levels, and the soil remains partially intact after failure due to root interconnection, thereby demonstrating greater toughness and integrity [43]. However, overly dense roots weaken the reinforcing effect of individual roots and increase pore irregularity, increasing the compressible space within the soil. As a result, the axial strain corresponding to the peak increases again, though it remains lower than that of the bare soil specimen. When the roots are too sparse, their improvement of the soil's pore structure is limited, they cannot provide additional cohesion, and their influence region is restricted, resulting in a smaller increase in strength.

The stress–strain curves obtained from numerical simulations also exhibit strain-softening behavior, and agree well with the experimental results within the pre-peak and near-peak axial strain range (about 0–3%). The simulated peak values are slightly lower than the experimental results, with the discrepancies within 10%. This is because numerical models cannot fully reproduce the pore structure of soil, and they also simplify conditions such as skeleton roughness and root–soil interface adhesion. The root reinforcement effect is represented by modified strength parameters rather than explicit root geometry and root–soil

interaction, while FEM under large deformation is also affected by deformation localization and element distortion [26, 31]. Therefore, the model can capture the overall strengthening effect of roots, but its ability to reproduce the detailed post-peak deformation and failure process is limited. Overall, the current numerical simulation results match the experimental results well and can reflect the mechanical behavior of the root reinforcement of lateritic soil.

The stress and displacement fields obtained from numerical simulations are shown in Fig. 21. Under the bare soil condition, the Von Mises stress exhibits a strip-like distribution under peak loading, with significant stress concentrations that tend to form a continuous failure surface. This behavior is primarily influenced by the pore distribution characteristics [60]. As seeding density increases, the pore distribution becomes more uniform, resulting in a more uniform stress field and a reduction in stress concentration effects [21]. However, under excessively high seeding density, localized high-stress channels reappear, with stress concentration regions forming around macropores.

The displacement field shows that the cumulative displacement at the soil surface is the greatest and decreases gradually downward layer by layer. In the bare soil model, overall deformation is more pronounced, with a large region of high displacement, primarily due to its higher porosity. In the grassed models, the overall porosity is lower, and the porosity is more uniformly distributed across elements, leading to reduced deformation at the top under axial loading. Additionally, since the axial load was applied through a rigid top plate, the displacement is distributed relatively evenly across the upper surface. However, the variation in element-wise porosity, combined with weaker lateral constraints at the edges, leads to stress redistribution during loading. This results in greater deformation near the model edges and less deformation near the center, forming an arched region of low deformation at the bottom of the model. After grass planting, the low deformation regions expand and the arched distribution becomes more distinct, showing improved stress dispersion and enhanced resistance to deformation.

4 Discussions

The reinforcement effect of vegetation on soil is largely governed by seeding density, which modifies the soil pore structure through root growth and development. By integrating CT scanning with numerical simulations, this study quantitatively evaluated the influence of ryegrass roots at different seeding densities on the pore structure and hydro-mechanical behavior of lateritic soil. The results showed

that a moderate seeding density (15 g/m^2) yielded the most favorable improvement in both pore structure and hydro-mechanical performance. This finding provides a scientific basis for vegetation-based slope protection design and supports earlier hypotheses, suggesting the existence of an optimal root content for maximizing reinforcement efficiency.

At moderate seeding density, the ryegrass roots effectively restructured the pore network of lateritic soil. Under this condition, the soil exhibited reduced total porosity (Fig. 11a) while increased tortuosity of pore channels (Fig. 16b), which enhanced flow resistance and lowered the saturated hydraulic conductivity (Fig. 19). This mechanism helped delay the buildup of pore water pressure during rainfall infiltration. These observations are consistent with previous findings that plant roots can slow down water infiltration by blocking flow paths or actively absorbing water [9, 25, 56]. This is further supported by Wang et al. [50], who observed via microtomography that root development significantly influences local pore distribution and leads to a global decrease in saturation. However, some studies have reported that overly dense or decayed roots may form interconnected macropores and induce preferential flow [8, 33]. The ryegrass in this study was at an active growth stage, dominated by fine roots, and no decaying roots were observed. These fine roots reduced soil permeability by refining the pore structure, rather than generating distinct preferential seepage paths as observed in decayed roots.

Ryegrass roots at moderate seeding density also showed remarkable mechanical reinforcement effects, significantly enhancing the strength and deformation resistance of lateritic soil. This improvement can be attributed to four primary mechanisms. (i) Roots divide the existing macropores to minimize pore size inhomogeneity, reducing potential weak surfaces and stress concentrations. (ii) Roots anchor the soil on both sides of the cracks to inhibit crack expansion, improving soil cohesion and resistance to deformation [24, 40]. (iii) Roots promote soil particle aggregation through secretions [12, 25], which improves structural stability. (iv) Roots possess a strong water absorption capacity, which changes the water distribution in the soil and increases the matric suction in localized zones of the soil, thus increasing the soil strength [28, 39, 67]. As a result of these mechanisms, the planted soil exhibited substantially higher strength than bare soil. This interpretation is also supported by previous studies showing that micro/meso-scale morphological and packing descriptors can influence soil stiffness and mechanical response through their effects on internal structure and force transmission [19, 54]. In this study, the changes in porosity, fractal dimension, pore morphology, coordination number, and tortuosity together reflect the structural basis

of the observed differences in deformation resistance and strength among specimens with different seeding densities. However, the behavior observed in this study is different from the continuous strength increase that is often reported for sands [22]. In sandy soils, increasing root volume density usually leads to a steady gain in strength because there is no strong aggregated structure to be disturbed. In lateritic soils, when the seeding density was excessively high, root entanglement disrupted some of the aggregates and the formation of additional large pores, slightly reducing the soil's deformation resistance. This finding indicates that beyond a certain vegetation density, further increases in root content may not necessarily provide additional mechanical benefits to lateritic soils.

In summary, a moderate seeding density enabled simultaneous improvement of both permeability resistance and strength of lateritic soils in the early stage of ryegrass growth, avoiding the misunderstanding that “more vegetation is always better.” Among the tested cases, the specimen with a seeding density of 15.0 g/m^2 showed the best overall performance, suggesting that moderate planting is more suitable than either sparse or excessive planting for ecological protection of lateritic soil slopes. This interpretation is supported not only by the CT-derived pore parameters, but also by the replicated permeability tests and the numerical analysis. Unlike previous studies that mainly compared the presence or absence of vegetation, this work emphasizes the importance of optimizing seeding density and provides practical guidance for vegetation-based slope stabilization.

It should be noted that, although all specimens were compacted from the same sieved lateritic soil and prepared under identical initial dry density, water content, geometry, and growth management, minor differences in the initial pore arrangement among specimens cannot be completely avoided. However, these differences are relatively small compared with the pore restructuring induced by root growth under different seeding densities. The findings of this study were obtained under a 28-day planting period and a specific soil–plant combination (lateritic soil and ryegrass). The proposed optimal seeding density and corresponding reinforcement effects are thus applicable primarily to early-stage lateritic soil-ryegrass reinforcement systems. Nevertheless, the general trends identified in this study are expected to be broadly applicable, while the optimal seeding density may vary depending on soil types, plant species, or long-term field conditions. In addition, the variation in actual root growth direction was not explicitly considered in this study, which may affect the quantitative estimation of the additional root cohesion.

5 Conclusions

This study systematically examined the effects of seeding densities of ryegrass on the pore structure and hydro-mechanical behavior of lateritic soil using CT scanning and numerical simulations. An optimal seeding density during the early growth period was identified, which can simultaneously enhance both the hydraulic and mechanical behavior of lateritic soil. The main conclusions are as follows:

1. Root mass and bulk parameters increased significantly with seeding density, while water absorption capacity remained relatively stable. A moderate seeding density (15.0 g/m^2) promoted a uniform distribution of root length and diameter. In contrast, an excessive seeding density (60.0 g/m^2) limited the growth of most roots due to increased root competition for water and nutrients, with only a few coarse and long roots.
2. With increasing seeding density, the total porosity and fractal dimension of lateritic soil decreased, and the pores became finer and more evenly distributed. At a high seeding density, some roots entangle with each other, causing macropores to reappear in the soil, which slightly increased soil porosity and fractal dimension. This suggests that the improvement of pore structure is weakened under dense root conditions.
3. At a moderate density (15.0 g/m^2), pore connectivity was weakest and seepage paths were most tortuous, leading to the lowest hydraulic conductivity. With further increases in seeding density, hydraulic conductivity slightly increased due to the reappearance of primary seepage paths (e.g., macropores).
4. Ryegrass roots increased the UCS and stiffness of lateritic soil by inhibiting the formation of cracks and macropores as well as homogenizing the stress field. However, local macropores formed by root entanglement promoted stress concentration, which induced internal damage and subsequently reduced the soil stiffness and deformation resistance.
5. During the early growth period of ryegrass, a seeding density of 15.0 g/m^2 achieved an optimal balance of infiltration control and mechanical reinforcement. From an engineering perspective for lateritic soil slopes, it is recommended that ryegrass seeding is maintained within a moderate density range to optimize both permeability reduction and strength enhancement in slope protection projects.

The research results provide a quantitative basis for optimizing ryegrass seeding density in ecological slope protection during the early growth period. Future studies should consider longer growth periods, diverse root

architectures, and the long-term effects of root senescence and decay, and in situ monitoring combined with coupled multi-field simulations, to enhance the generalizability and applicability of the findings.

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Author contribution Hui-Cong Yu helped in conceptualization, writing—original draft, data curation, investigation. Ling Zeng contributed to conceptualization, supervision, funding acquisition. Jing Tang helped in methodology and investigation. Jing-Cheng Chen and Jin-Tao Luo validated the study. Xin-Yang Wu helped in investigation and data curation. Qian-Feng Gao contributed to supervision and writing—review & editing. Han-Bing Bian helped in methodology, formal analysis, writing—review & editing.

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Data availability The data supporting the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

References

1. Bordoloi S, Ng CWW (2020) The effects of vegetation traits and their stability functions in bio-engineered slopes: a perspective review. *Eng Geol* 275:105742. <https://doi.org/10.1016/j.enggeo.2020.105742>
2. Bordoloi S, Ni J, Ng CWW (2020) Soil desiccation cracking and its characterization in vegetated soil: a perspective review. *Sci Total Environ* 729:138760. <https://doi.org/10.1016/j.scitotenv.2020.138760>
3. Cai H, Zhang Q, Ye G, Lu L, Wu C, Ji F (2023) Investigation on strong vertical permeability and micro-pore characteristics of Jinan red clay using X-ray micro-tomography. *Environ Earth Sci* 82:122. <https://doi.org/10.1007/s12665-023-10810-9>
4. Cai P, Que Y, Jiang Z, Li X (2022) 3D quantitative characterization and flow simulation of granite residual soil based on CT scanning. *Sci Sin Technol* 52:1065–1082. <https://doi.org/10.1360/SST-2021-0308>
5. Cheng Q, Gu YD, Tang CS, Zhang XY, Shi B (2024) Desiccation cracking behaviour of a vegetated soil incorporating planting density. *Can Geotech J* 61:165–173. <https://doi.org/10.1139/cgj-2023-0047>

6. Consoli NC, Párraga Morales D, Saldanha RB (2021) A new approach for stabilization of lateritic soil with Portland cement and sand: strength and durability. *Acta Geotech* 16:1473–1486. <https://doi.org/10.1007/s11440-020-01136-y>
7. Cruz BC, Furrer JM, Guo YS, Dougherty D, Hinestroza HF, Hernandez JS, Gage DJ, Cho YK, Shor LM (2017) Pore-scale water dynamics during drying and the impacts of structure and surface wettability. *Water Resour Res* 53:5585–5600. <https://doi.org/10.1002/2016WR019862>
8. Cui Z, Huang Z, Liu Y, López-Vicente M, Wu GL (2022) Natural compensation mechanism of soil water infiltration through decayed roots in semi-arid vegetation species. *Sci Total Environ* 819:151985. <https://doi.org/10.1016/j.scitotenv.2021.151985>
9. Deng Z, Lan H, Li L, Sun W (2025) Vegetation-induced modifications in hydrological processes and the consequential dynamic effects of slope stability. *CATENA* 251:108793. <https://doi.org/10.1016/j.catena.2025.108793>
10. Dong H, Luo X, Li ZF (2017) Quantitative influence of mesoporosity space features of aggregate gravel soil on its permeability characteristics. *J Cent South Univ (Sci Technol)* 48:1367–1375. <https://doi.org/10.11817/j.issn.1672-7207.2017.05.032>
11. Emeka OJ, Nahazanan H, Kalantar B, Khuzaimah Z, Sani OS (2021) Evaluation of the effect of hydroseeded vegetation for slope reinforcement. *Land* 10:995. <https://doi.org/10.3390/land10100995>
12. Fraccica A, Romero E, Fourcaud T (2025) Effects of vegetation growth on soil microstructure and hydro-mechanical behaviour. *Géotechnique* 75:293–307. <https://doi.org/10.1680/jgeot.23.00163>
13. Fu H, Zha H, Zeng L, Chen C, Jia C, Bian H (2020) Research progress on ecological protection technology of highway slope: status and challenges. *Transp Saf Environ* 2:3–17. <https://doi.org/10.1093/tse/tdaa006>
14. Gao QF, Wu XY, Zeng L, Yu HC, Shi XK (2025) Reliability analysis of soil slope stability considering spatial variability of desiccation cracks. *Comput Geotech* 179:106977. <https://doi.org/10.1016/j.compgeo.2024.106977>
15. Gao QF, Yu HC, Zeng L, Huang YX (2023) Characterization of water retention behavior of cracked compacted lateritic soil exposed to wet-dry cycles. *Bull Eng Geol Environ* 82:61. <https://doi.org/10.1007/s10064-023-03089-4>
16. Gao QF, Wu XY, Zeng L, Yu HC, Yu H (2024) Mechanisms behind wetting self-healing behavior and its influence on the strength of lateritic soil. *China J Highw Transp* 37:157–168
17. Giménez D, Allmaras RR, Huggins DR, Nater EA (1997) Prediction of the saturated hydraulic conductivity-porosity dependence using fractals. *Soil Sci Soc Am J* 61:1285–1292. <https://doi.org/10.2136/sssaj1997.03615995006100050001x>
18. Giuliani LM, Hallett PD, Loades KW (2024) Effects of soil structure complexity to root growth of plants with contrasting root architecture. *Soil Tillage Res* 238:106023. <https://doi.org/10.1016/j.still.2024.106023>
19. Gong J, Pang X, Tang Y, Liu M, Jiang J, Ou X (2024) Effects of particle shape, physical properties and particle size distribution on the small-strain stiffness of granular materials: a DEM study. *Comput Geotech* 165:105903. <https://doi.org/10.1016/j.compgeo.2023.105903>
20. Hallett PD, Dexter AR, Seville JPK (1995) Identification of pre-existing cracks on soil fracture surfaces using dye. *Soil Tillage Res* 33:163–184. [https://doi.org/10.1016/0167-1987\(94\)00477-V](https://doi.org/10.1016/0167-1987(94)00477-V)
21. Huang J, Dai J, Scarpa F, Wang Y, Ji J, Mao Z (2025) Random root distribution affects the mechanical properties of the soil-root composite and root reinforcement. *CATENA* 255:108896. <https://doi.org/10.1016/j.catena.2025.108896>
22. Jiang YJ, Xia X, Hu XB, Zhu YJ (2025) Effect of root volume density on the mechanical behaviour of saturated sand under drained and undrained conditions. *Acta Geotech* 20:3845–3869. <https://doi.org/10.1007/s11440-025-02605-y>
23. Lang Y, Liang Z, Dong Z (2024) Three-dimensional finite element simulation and reconstruction of jointed rock models using CT scanning and photogrammetry. *J Rock Mech Geotech Eng* 16:1348–1361. <https://doi.org/10.1016/j.jrmge.2023.08.008>
24. Lann T, Bao H, Lan H, Zheng H, Yan C, Peng J (2024) Hydro-mechanical effects of vegetation on slope stability: a review. *Sci Total Environ* 926:171691. <https://doi.org/10.1016/j.scitotenv.2024.171691>
25. Li J, Cui P, Yin Y (2023) Field observation and micro-mechanism of roots-induced preferential flow by infiltration experiment and phase-field method. *J Hydrol* 623:129756. <https://doi.org/10.1016/j.jhydrol.2023.129756>
26. Liang Q, Lan H, Zhou Y, Li B, Sun W, Liu S, Lv W (2025) Reverse size effect of the unconfined compressive strength of crystalline rock: a grain-scale perspective. *Rock Mech Rock Eng* 58:1651–1669. <https://doi.org/10.1007/s00603-024-04216-7>
27. Luo Y, Wen T, Lin X, Chen X, Shao L (2024) Quantitative analysis of pore-size influence on granite residual soil permeability using CT scanning. *J Hydrol* 645:132133. <https://doi.org/10.1016/j.jhydrol.2024.132133>
28. Mahannopkul K, Jotisankasa A (2019) Influences of root concentration and suction on *Chrysopogon zizanioides* reinforcement of soil. *Soils Found* 59:500–516. <https://doi.org/10.1016/j.sandf.2018.12.014>
29. Mao Z, Ma X, Geng M, Wang M, Gao G, Tian Y (2025) Development characteristics and quantitative analysis of cracks in root-soil complex during different growth periods under dry-wet cycles. *Biogeotechnics* 3:100121. <https://doi.org/10.1016/j.bgtech.2024.100121>
30. Mao Z, Wang M, Xu G, Geng M, Ma X, Gao G, Tian Y, Wang L, Xi Y (2025) The coupled temporal effects and micro-mechanism of root reinforcement and dry-wet cycles on the strength of herb-loess composite. *Soil Tillage Res* 253:106684. <https://doi.org/10.1016/j.still.2025.106684>
31. Mardalizad A, Scazzosi R, Manes A, Giglio M (2018) Testing and numerical simulation of a medium strength rock material under unconfined compression loading. *J Rock Mech Geotech Eng* 10:197–211. <https://doi.org/10.1016/j.jrmge.2017.11.009>
32. Matocha CJ, Karathanasis TD, Murdock LW, Grove JH, Goodman J, Call D (2018) Influence of ryegrass on physico-chemical properties of a fragipan soil. *Geoderma* 317:32–38. <https://doi.org/10.1016/j.geoderma.2017.12.004>
33. Meng S, Zhang T, Zhao G, Hou S (2025) Time-varying mechanisms of hydraulic properties of root-soil composites under plant root decay. *J Hydrol* 658:133192. <https://doi.org/10.1016/j.jhydrol.2025.133192>
34. Ng CWW, Zhang Q, Guo H, Ni J, Wang Y, Leung AK, Zhou C (2024) Eco-geotechnics under climate change: a state-of-the-art review. *Biogeotechnics*. <https://doi.org/10.1016/j.bgtech.2024.100158>
35. Ni JJ, Leung AK, Ng CWW, Shao W (2018) Modelling hydro-mechanical reinforcements of plants to slope stability. *Comput Geotech* 95:99–109. <https://doi.org/10.1016/j.compgeo.2017.09.001>
36. Nong R, Wan Y, Ding Y, Chai M, Zhu L (2023) Effects of root systems on crack formation: experiments, modeling, and analyses. *Soil Tillage Res* 233:105784. <https://doi.org/10.1016/j.still.2023.105784>
37. Oliveira RCD, Pedroso CEDS, Soares VN, Tunes LMD, Meneghello GE, Rabelo CHS (2019) Sowing depth of annual ryegrass seed on different substrates as a vigor test. *J Seed Sci* 41:196–204. <https://doi.org/10.1590/2317-1545v41n2212421>
38. Ouyang M, Lan RY, Zhang HR, Wang GY, Qin GH, Guo O, Li CG (2025) Shear-strength model of root-soil interface

- considering interfacial suction and its influencing factors. *China J Highw Transp* 38:108–118
39. Pallewatttha M, Indraratna B, Heitor A, Rujikiatkamjorn C (2019) Shear strength of a vegetated soil incorporating both root reinforcement and suction. *Transp Geotech* 18:72–82. <https://doi.org/10.1016/j.trgeo.2018.11.005>
 40. Prasetyaningtiyas GA, Kamchoom V, Leung AK, Likitlersuang S (2024) Hydromechanical behaviour of a slope reinforced by grass roots under rainfall conditions. *Ecol Eng* 209:107427. <https://doi.org/10.1016/j.ecoleng.2024.107427>
 41. Que Y, Chen X, Jiang Z, Cai P, Xue B, Xie X (2024) Pore-scale permeability estimation of undisturbed granite residual soil: a comparison study by different methods. *J Soils Sediments* 24:1591–1613. <https://doi.org/10.1007/s11368-024-03745-6>
 42. Rabbani A, Babaei M (2019) Hybrid pore-network and Lattice-Boltzmann permeability modelling accelerated by machine learning. *Adv Water Resour* 126:116–128. <https://doi.org/10.1016/j.advwatres.2019.02.012>
 43. Song B, Nakamura D, Kawaguchi T, Kawajiri S, Rui D (2025) Quantifying the shear behavior of fine-grained soil with herbaceous plant roots under freeze-thaw conditions using X-ray CT scan. *Soil Tillage Res* 246:106326. <https://doi.org/10.1016/j.still.2024.106326>
 44. Song R, Wu M, Wang Y, Liu J, Yang C (2023) In-situ X-CT scanning and numerical modeling on the mechanical behavior of the 3D printing rock. *Powder Technol* 416:118240. <https://doi.org/10.1016/j.powtec.2023.118240>
 45. Soto-Gómez D, Pérez-Rodríguez P, Vázquez-Juiz L, López-Periago JE, Paradelo M (2018) Linking pore network characteristics extracted from CT images to the transport of solute and colloid tracers in soils under different tillage managements. *Soil Tillage Res* 177:145–154. <https://doi.org/10.1016/j.still.2017.12.007>
 46. Sun H, Gao Y, Zheng X, Chen Y, Jiang Z, Zhang Z (2019) Meso-scale simulation of concrete uniaxial behavior based on numerical modeling of CT images. *Materials* 12:3403. <https://doi.org/10.3390/ma12203403>
 47. Tan S, Xiang G, Xu X, Liu T (2025) Mechanical characteristics of herbaceous plant root system and slope stability research. *Sci Rep* 15:24916. <https://doi.org/10.1038/s41598-025-09581-z>
 48. Tan X, Huang Y, Xiong D, Lv K, Chen F (2020) The effect of *Elymus nutans* sowing density on soil reinforcement and slope stabilization properties of vegetation–concrete structures. *Sci Rep* 10:20462. <https://doi.org/10.1038/s41598-020-77407-1>
 49. Waldron LJ (1977) The shear resistance of root-permeated homogeneous and stratified soil. *Soil Sci Soc Am J* 41:843–849. <https://doi.org/10.2136/sssaj1977.03615995004100050005x>
 50. Wang JP, Sha JF, Gao XG, Dadda A, Qi M, Ge S (2024) Experimental investigation of the microscopic interaction mechanism between sand and tall fescue roots by using X-ray microtomography. *Acta Geotech* 19:1529–1554. <https://doi.org/10.1007/s11440-024-02234-x>
 51. Wang Q, Milatz M, Hosseini R, Kumar K (2023) Multiphase lattice Boltzmann modeling of cyclic water retention behavior in unsaturated sand based on X-ray computed tomography. *Can Geotech J* 60:1429–1446. <https://doi.org/10.1139/cgj-2022-0489>
 52. Wen T, Luo Y, Tang M, Chen X, Shao L (2024) Effects of representative elementary volume size on three-dimensional pore characteristics for modified granite residual soil. *J Hydrol* 643:132006. <https://doi.org/10.1016/j.jhydrol.2024.132006>
 53. Wu TH, McKinnell Iii WP, Swanson DN (1979) Strength of tree roots and landslides on Prince of Wales Island, Alaska. *Can Geotech J* 16:19–33. <https://doi.org/10.1139/t79-003>
 54. Xiao Y, Liang F, Zhang Z, Cui H, Liu H (2025) Thermodynamic model for sand-incorporating morphology. *Int J Geomech* 25:04025041. <https://doi.org/10.1061/IJGNAI.GMENG-10598>
 55. Yu HC, Zeng L, Wu XY, Gao QF, Bian HB, Luo JT, Chen JC, Zhang HR (2025) Classification of cracking potential for clayey soils based on cyclic wet-dry tests. *Environ Earth Sci* 84:198. <https://doi.org/10.1007/s12665-025-12210-7>
 56. Yuliana Y, Apriyono A, Kamchoom V, Boldrin D, Cheng Q, Tang CS (2025) Seasonal dynamics of root growth and desiccation cracks and their effects on soil hydraulic conductivity. *Eng Geol* 349:107973. <https://doi.org/10.1016/j.enggeo.2025.107973>
 57. Zeng L, Li F, Liu J, Gao Q, Bian H (2019) Effect of initial gravimetric water content and cyclic wetting-drying on soil-water characteristic curves of disintegrated carbonaceous mudstone. *Transp Saf Environ* 1:230–240. <https://doi.org/10.1093/tse/tdz018>
 58. Zeng L, Yu H, Gao Q, Bian H (2020) Mechanical behavior and microstructural mechanism of improved disintegrated carbonaceous mudstone. *J Cent South Univ* 27:1992–2002. <https://doi.org/10.1007/s11771-020-4425-8>
 59. Zeng L, Yu HC, Gao QF, Zhang YH, Bian HB (2023) Characterization of volumetric crack rate and its influence on unconfined compressive strength of red clay. *J Mater Civ Eng* 35:4023418. <https://doi.org/10.1061/JMCEE7.MTENG-15852>
 60. Zeng L, Yu HC, Qiu J, Luo JT, Liu J, Gao QF, Zhang HR (2024) Pore characteristics and microscopic damage mechanism of disintegrated carbonaceous mudstone exposed to dry-wet cycles. *Constr Build Mater* 433:136774. <https://doi.org/10.1016/j.conbuildmat.2024.136774>
 61. Zeng L, Yu HC, Yuan YR, Gao QF, Bian HB (2023) Evolution of seepage and stability characteristics of lateritic soil slopes containing a tension crack zone under rainfall infiltration. *Iran J Sci Technol Trans Civ Eng* 47:3717–3730. <https://doi.org/10.1007/s40996-023-01159-x>
 62. Zhang L, Wang J (2023) Prediction of the soil saturated hydraulic conductivity in a mining area based on CT scanning technology. *J Clean Prod* 383:135364. <https://doi.org/10.1016/j.jclepro.2022.135364>
 63. Zhang Z, Tian A, Zhou Y, Ding C, Tang Q (2025) Ryegrass root-soil composites mechanical properties and its slope stability: experimental study and numerical analysis. *J Air Waste Manag Assoc* 75:164–179. <https://doi.org/10.1080/10962247.2024.2436531>
 64. Zhao Y, Xu L, Qin C, Huang X, Wu Y (2025) Tracking the evolution of loess microstructure using micro-CT 3D reconstruction based on a soil-particle-aware model. *Comput Geotech* 182:107176. <https://doi.org/10.1016/j.compgeo.2025.107176>
 65. Zhou ZC, Shangguan ZP (2007) The effects of ryegrass roots and shoots on loess erosion under simulated rainfall. *CATENA* 70:350–355. <https://doi.org/10.1016/j.catena.2006.11.002>
 66. Zhu H, Zhang LM (2016) Field investigation of erosion resistance of common grass species for soil bioengineering in Hong Kong. *Acta Geotech* 11:1047–1059. <https://doi.org/10.1007/s11440-015-0408-6>
 67. Zhu H, Zhang LM, Garg A (2018) Investigating plant transpiration-induced soil suction affected by root morphology and root depth. *Comput Geotech* 103:26–31. <https://doi.org/10.1016/j.compgeo.2018.06.019>

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